From many origins to single events

An AI-run study of the genomic transitions to multicellularity

AI hackathon · [dates: ____] · two rounds, about 31 hours of agent work



A week-long test of agentic bioinformatics

Beltrao lab
ETH Zürich

Colm Ryan lab
UCD

Emanuel Gonçalves lab
IST

Christophe Dessimoz lab
UNIL

[dates: ____]
[__ participants] · 13 projects

Blanks to be filled by the organisers.

Human writes a 'predict X' prompt

Claude Opus 5.5 leads as PI

Delegates to Opus / Sonnet subagents

Builds, benchmarks, writes up

Humans verify afterwards

This project: own cluster, Singularity in /scratch · R1: 24 CPU, 150 GB · R2: 48 CPU, 500 GB, 1 A100

The hackathon in one picture

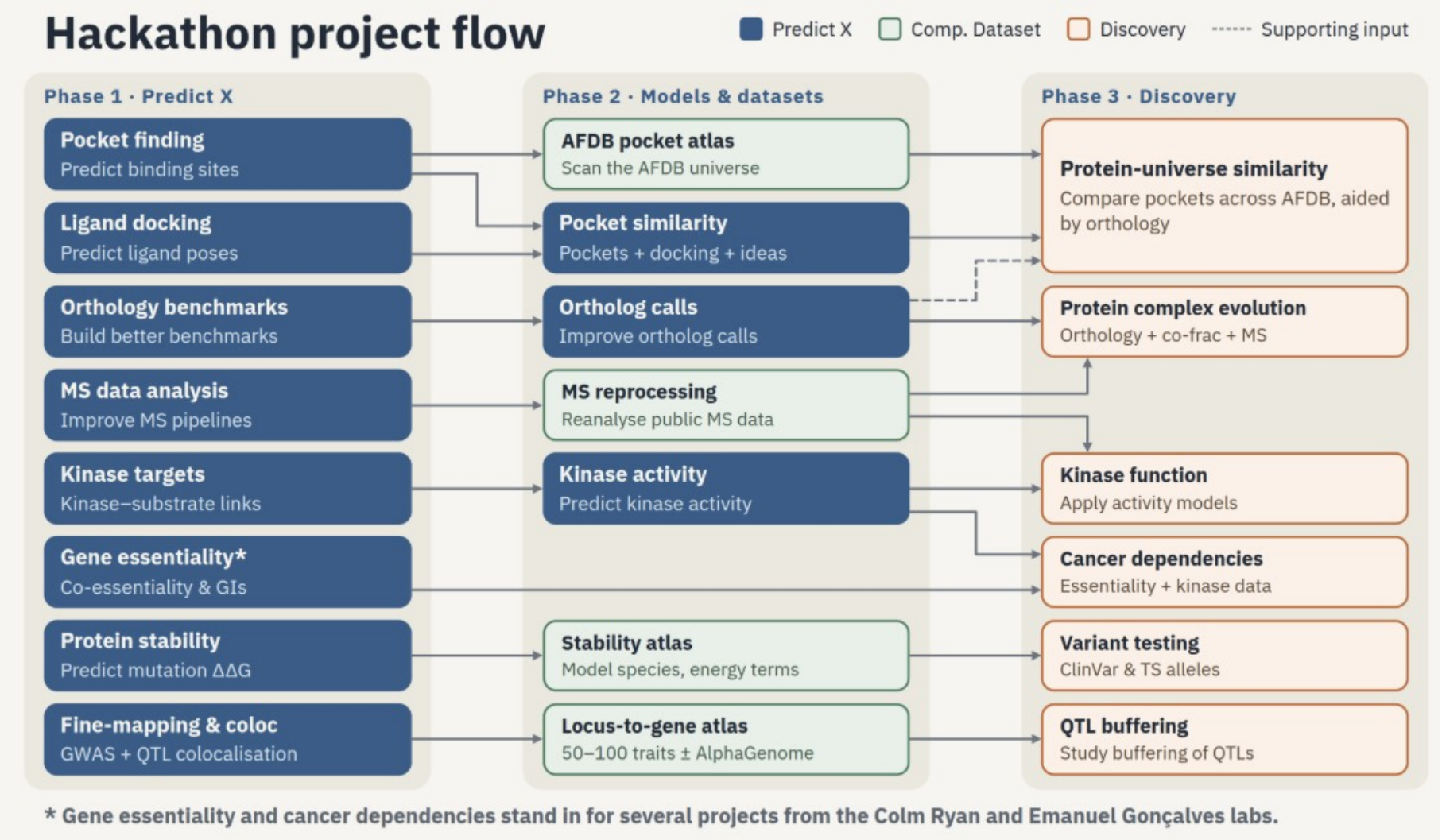
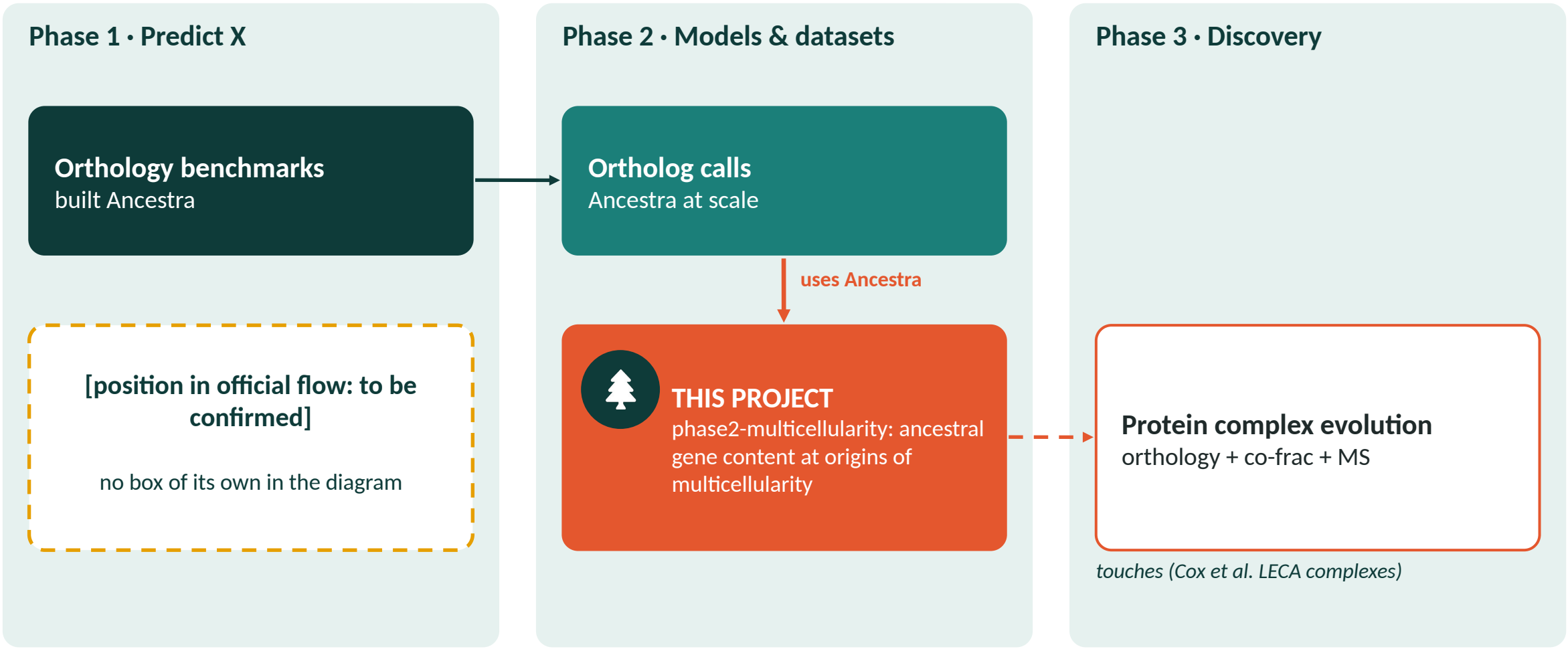


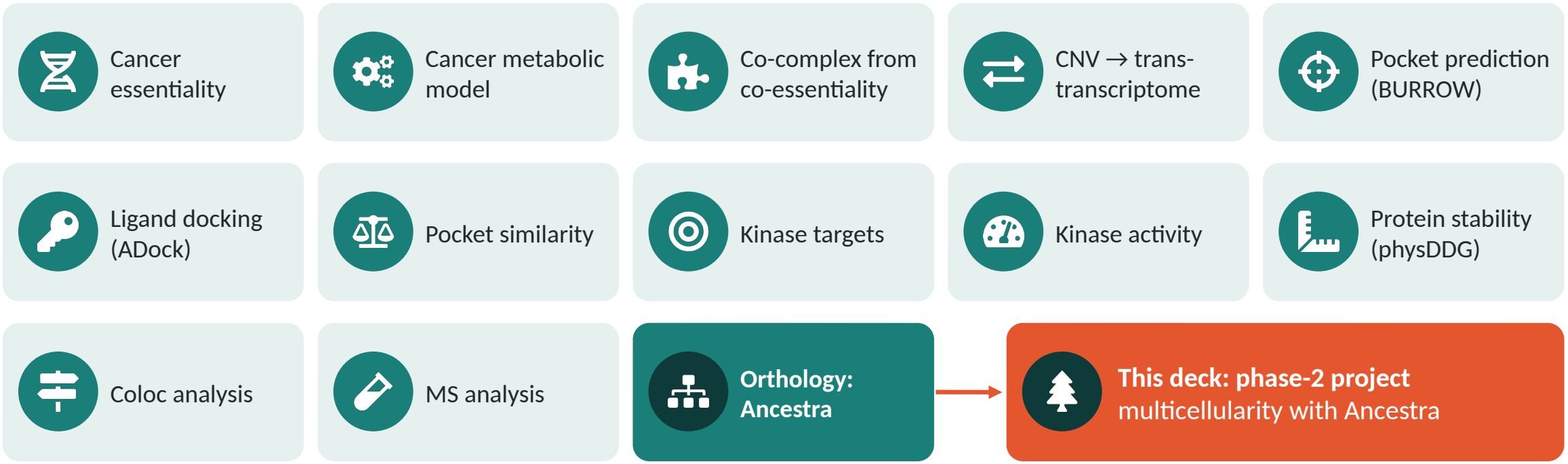
Image: hackathon organisers.

Where this project sits in the flow



Simplified from the organisers' flow diagram; only the boxes this project connects to are shown.

Thirteen projects, one shared recipe



Same recipe for all: one prompt, Claude as PI, subagents, human verification after the fact.

Every report opens with a human disclaimer

> *This is an exercise in assessing the bioinformatics capabilities of agentic AI … All results in this project were generated entirely by AI with minimal human intervention.*
>
> ***Currently none of the results have been verified by a human.***

Shortened from the hackathon's human-written header.

Generated by AI
minimal human intervention

Not a publication
an exercise in agentic science

Not yet verified
human checks pending

B

The initial prompt

What the human asked for, which parts are generic, and what changed in round 2



Anatomy of the round-1 prompt

GENERAL · reusable

- Role: PI agent who delegates
- Time budget + milestones
- Hardware and sandbox
- Literature + ≤3 design critiques
- Reproducibility rules
- Research tree
- AI lessons file
- Reporting + writing standards
- Human disclaimer



PROMPT.md
≈1,470 words

9 generic blocks

5 science blocks

Most of the prompt is a reusable research protocol; the science fits in one paragraph.

PROJECT-SPECIFIC

- **Kinds of multicellularity**
- **Ancestra: ancestral genomes**
- **EukProt, LukProt, 'Eukaryoma'**
- **Menu of analyses**
- **Selection-pressure question**

Round 2: seven new scientific asks

New resources: 48 CPU · 500 GB · one A100 GPU · ~30 h — and a department-head review first

1

EukaryOMA
the database round 1 'could not find'

2

3D structure
ProstT5 structure letters on GPU

3

Simple → complex
not only origins of multicellularity

4

One run per event
dense sampling around each origin

5

Motifs, death, TFs
motif-binding, cell-death, cell-type TF genes

6

Specific Pfam families
domain architectures, functions

7

Protein complexes
co-fractionation map (Marcotte lab)

Head review
Fable agent reads all of it, published

An external review set the round-2 agenda

What was wrong

- 7 heterogeneous origins averaged: 13–35% power
- Validation overstated: deep-node controls 0 of 3
- EukaryOMA wrongly 'not found'

What was good

- Hashed pre-registration, calibrated null
- Diagnosis of why controls failed
- Lead: novelty at simple → complex

What to do next

- Cross-check with EukaryOMA
- Dense per-event Ancestra runs
- Simple → complex, specific families

Fable agent as department head · reviews/round2_department_head_review.txt · public page linked at the end

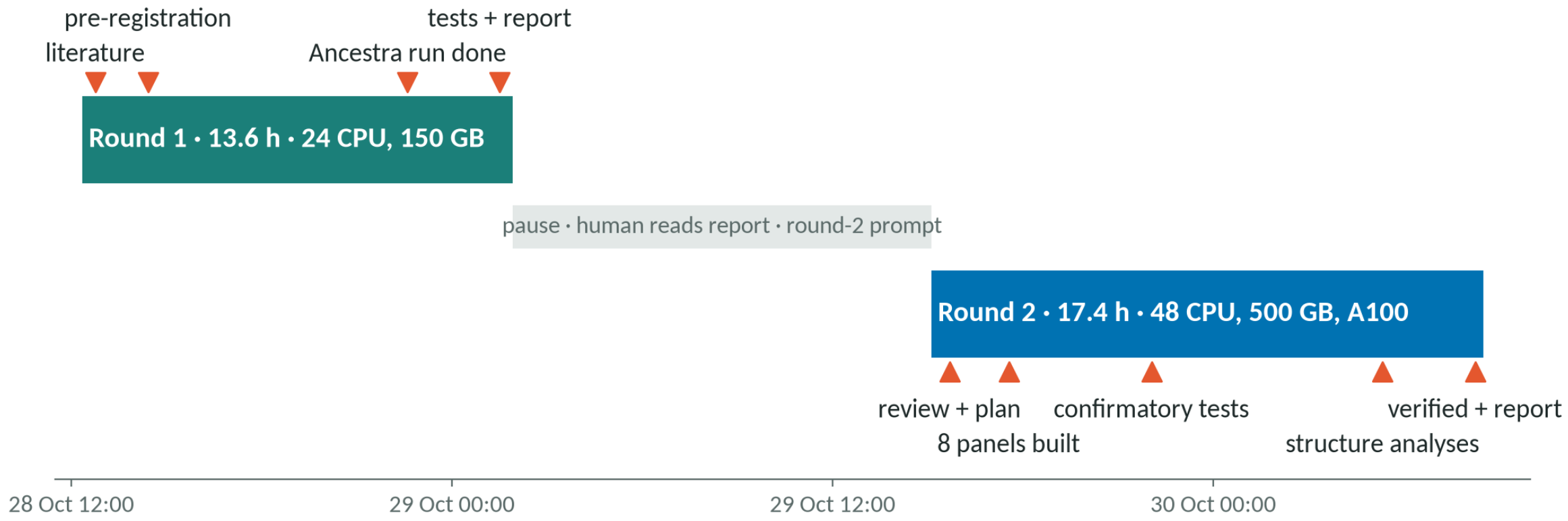
C

Overview

Two rounds, one AI team, about 31 hours

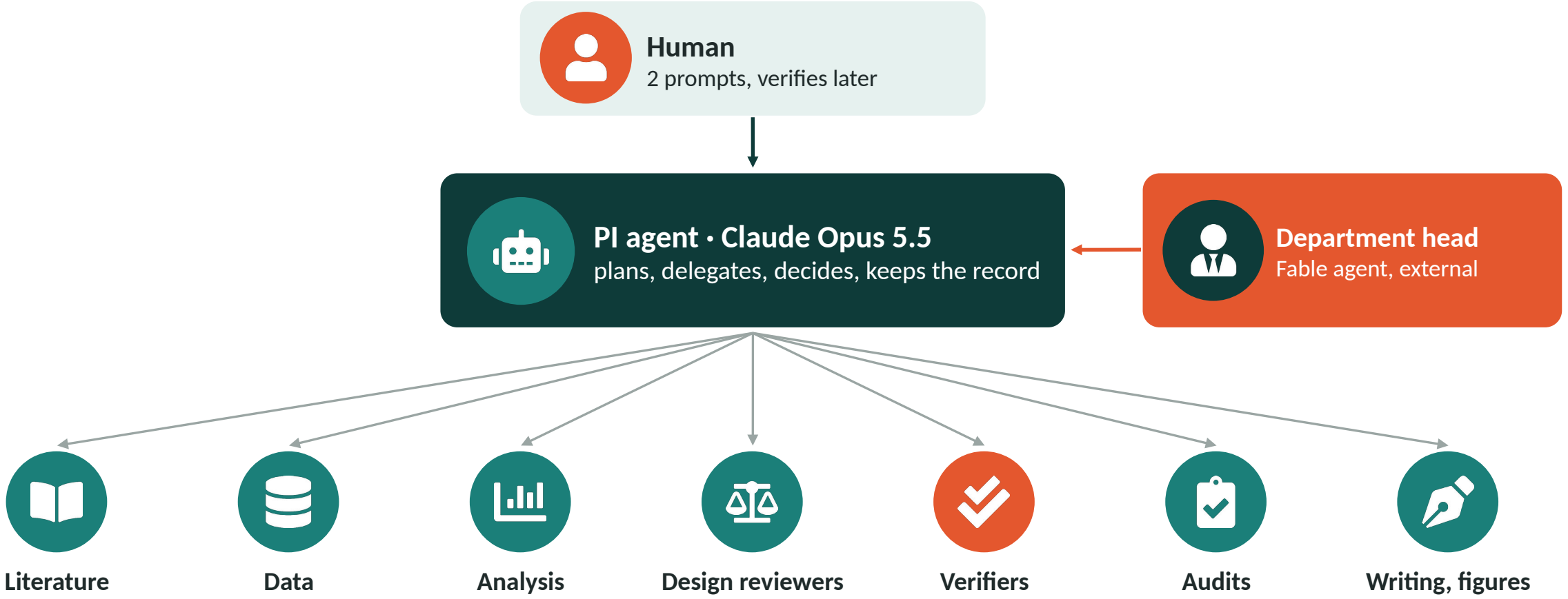


Two rounds in about 31 hours



Milestones from RESEARCH_TREE.md. Wall-clock time; the agent team worked in parallel inside each round.

One PI agent, many specialised agents



Subagents: Opus 5.5 or Sonnet 5.5, chosen by the PI · verifiers never checked their own work

The experiment in numbers

187 + 82
eukaryotic + prokaryotic proteomes (round 1)

2.97 M
eukaryotic proteins in round 1

8 · 227
dense panels · proteomes (round 2)

2 + 9
Ancestra runs, round 1 + round 2

16
literature hypotheses (~250 references)

3
frozen pre-registrations (git tags)

2
own claims overturned by verification

≈31 h
total wall time, two rounds

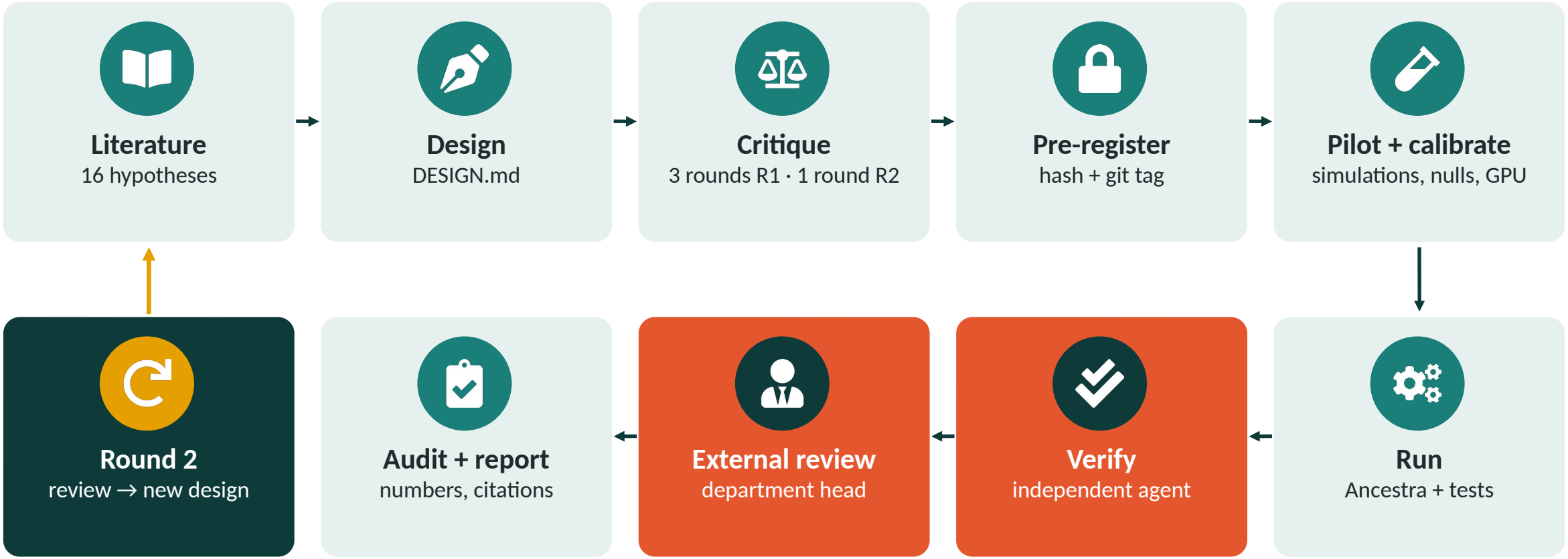
D

How the method was decided

Literature, critique, pre-registration, calibration, verification



A pipeline of checks before any claim



Red: steps done by agents that did not produce the result.

Key decisions, round 1

1 · Separate runs

Observed Orthology across bacteria and eukaryotes adds cost; no transition spans both

Decided Separate eukaryote and prokaryote Ancestra runs

Effect The 1.4 h prokaryote run became the dry run

2 · Novelty ladder

Observed Simulations: raw gene gains inflated 2.1–2.6×

Decided Count only sequence-novel gains; claim a burst only above 2.2×

Effect Rule fixed before any eukaryote output

3 · Calibrated null

Observed Planned null: 14.5% false positives at α = 0.05 on the real tree

Decided Calibrated permutation scheme as the primary test

Effect 4.9% false positives: p-values mean what they say

Key decisions, round 2

4 · Dense panels

Observed One average over 7 origins: 13–35% power; giant component of 73% of proteins

Decided 8 small per-event panels, fast preset, 15–45 min each

Effect Sharper events, but only 5–9 controls per panel

5 · Unseen events only

Observed Round-1 nested events already looked novel (p = 0.016)

Decided Confirmatory test on 4 new events, frozen before their runs

Effect No double dipping; power only 18–41%

6 · Rule before data

Observed Deep nodes were Ancestra's weak point

Decided Use structure mode only if controls improve and agreement holds

Effect Rule failed (23.3% → 20.7%): default kept

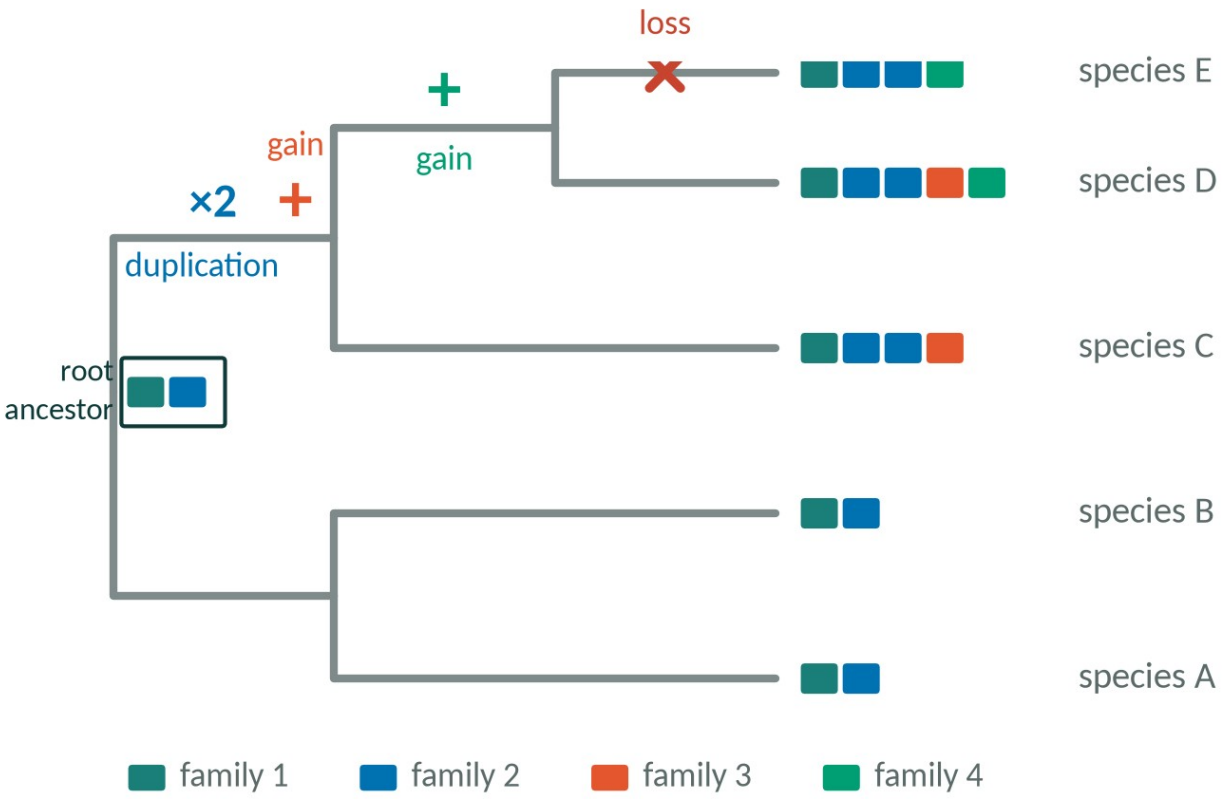
E

The method, explained

Simple pictures for the parts that non-experts find hard



What is ancestral gene content?



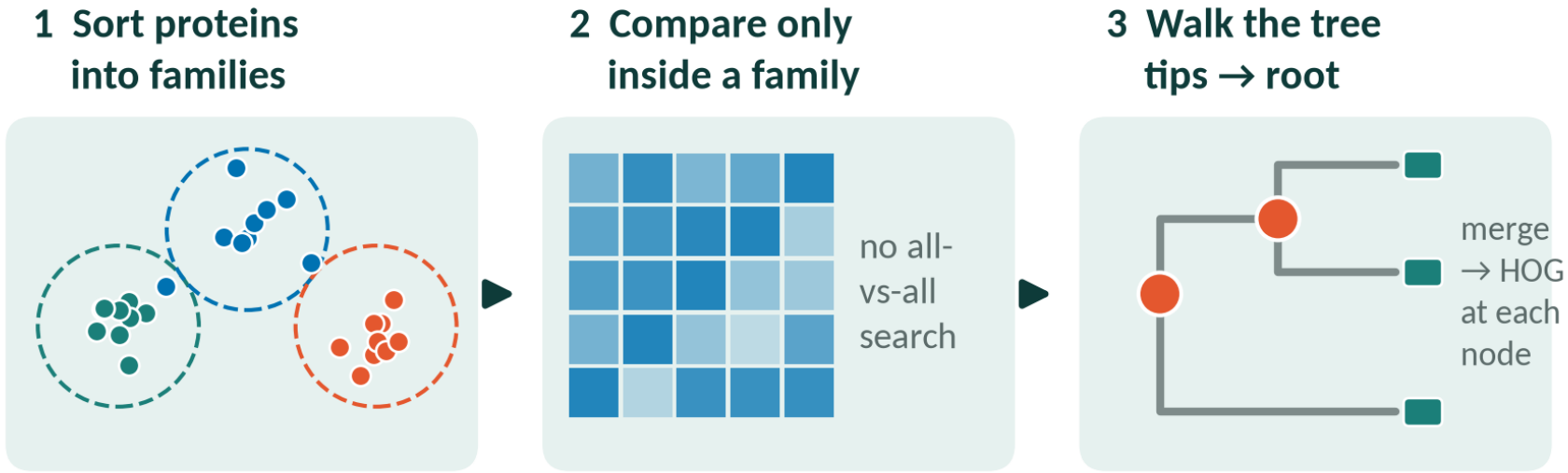
Gene family
genes descended from one ancestral gene

HOG
a family at one tree level = one ancestral gene

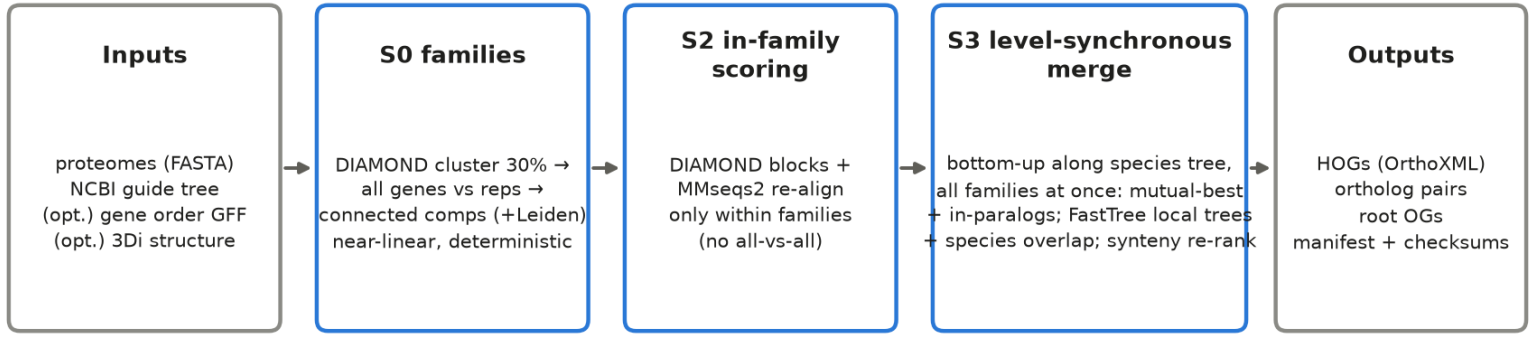
Events
gain, duplication or loss, read branch by branch

Schematic, not real data.

Ancestra in three steps



Ancestra: orthology as level-synchronous ancestral-genome reconstruction inside de novo families



Lower figure: Ancestra's own method overview, reused with credit (evocellnet/ancestra-mvp-ai, docs/figures/fig1_method.png).

Ancestra among orthology methods

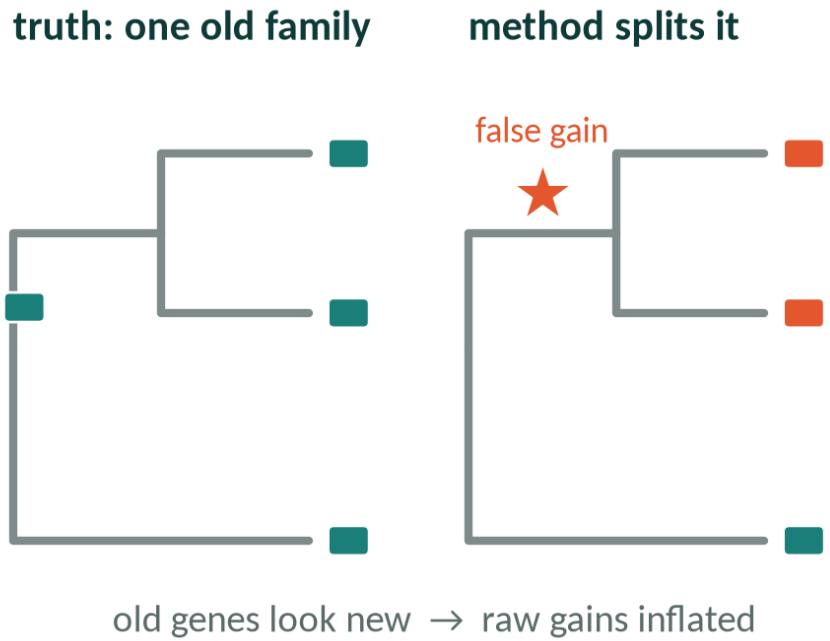
Method	How families are built	Output	OrthoBench F	QfO SwissTree recall
Ancestra (this project)	reference-free DIAMOND families, then level-by-level merge	HOGs at every tree node	76.2 / 74.8 (accurate / fast)	0.721 / 0.731
OrthoFinder 3.2	all-vs-all similarity graph + gene trees	orthogroups, gene trees	69.7	0.825
OMA · FastOMA (EukaryOMA)	pairwise orthologs; FastOMA places genes on a reference	HOGs at every level	60.4 / 33.1 ‡	0.696 / 0.699
SonicParanoid2 · InParanoid · Hieranoid	graph / pairwise; Hieranoid bottom-up	orthogroups, pairs, HOGs	70.4 (SonicParanoid2)	0.717 · 0.701 · 0.726
Pfam counts + Dollo / Count	domain annotation, no orthology	copy numbers on the tree	n/a	n/a

Shared homology from similarity graphs; a HOG = all genes from one ancestral gene

Different no reference database; all families merged level by level at once

Numbers from the Ancestra README. ‡ OMA comparators overlap the benchmark. Non-inferiority to OrthoFinder not shown; QfO recall below OrthoFinder.

Why 'new gene' counts are inflated



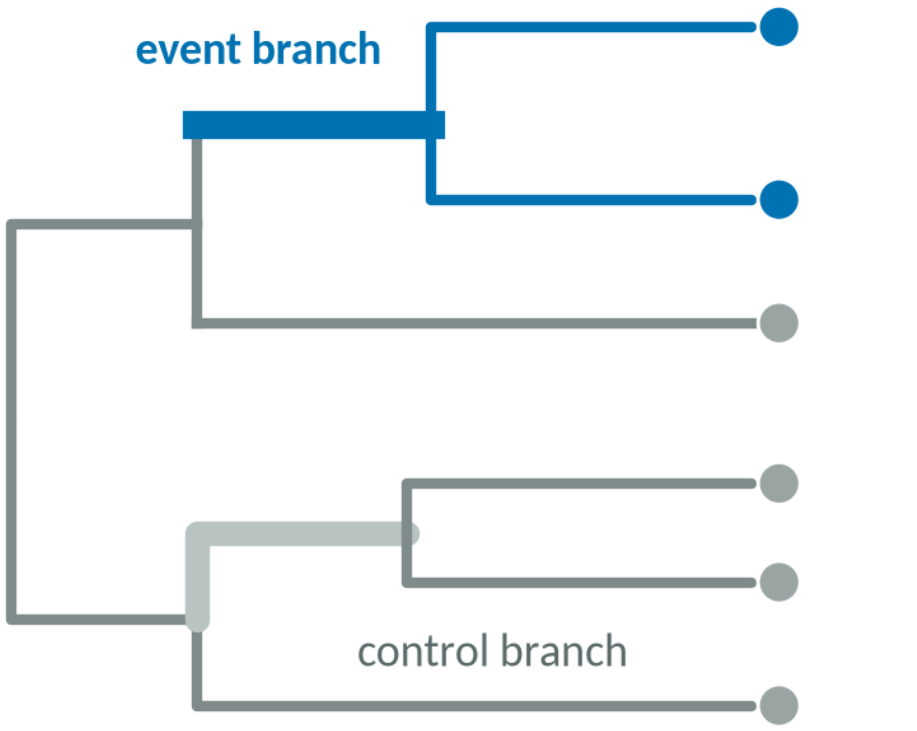
All HOG gains on a branch
raw Ancestra output; inflated 2.1–2.6× in simulations

Sequence-novel
no DIAMOND hit outside the clade (E ≤ 1e-5, ≥ 30% coverage)
41–80% of gains in the panels

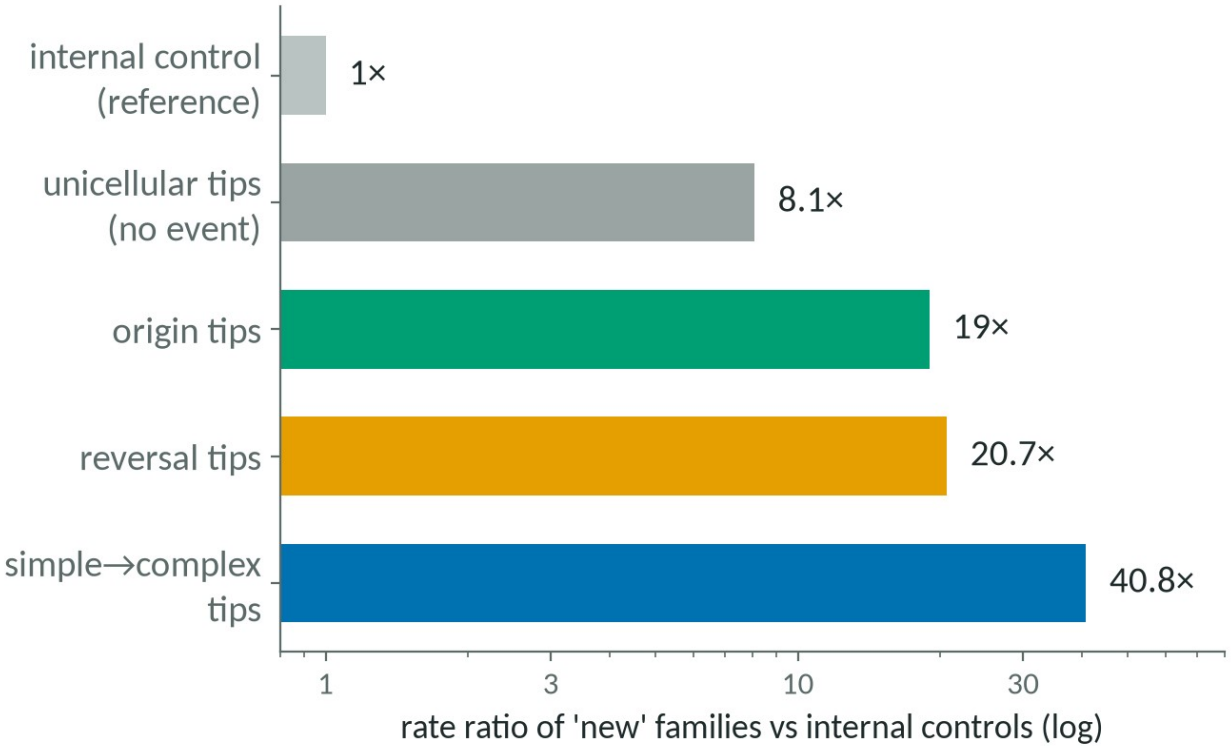
No structural match
Foldseek search: about half of sequence-novel families
have a structural relative outside the clade

Each rung removes 'new' genes that have detectable relatives outside the clade.

Compare like with like

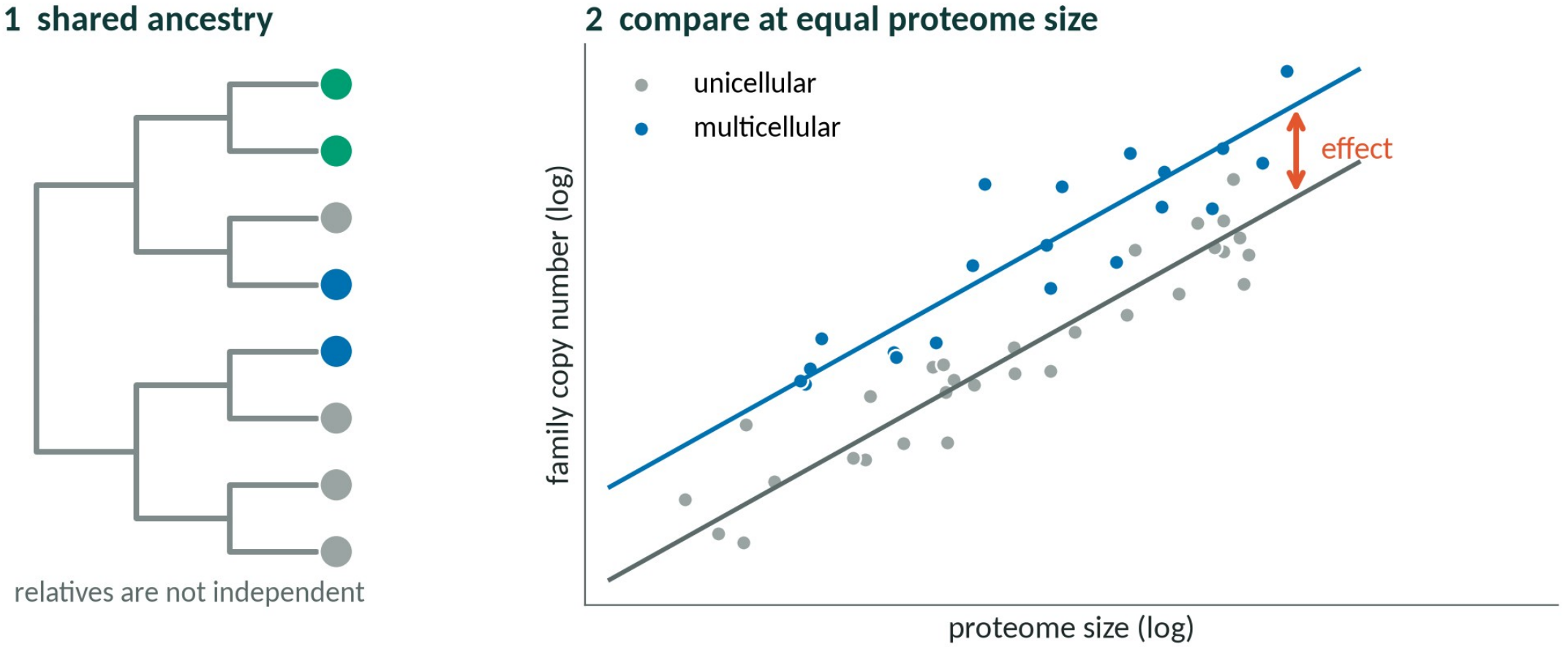


observed ÷ expected from controls = rate ratio



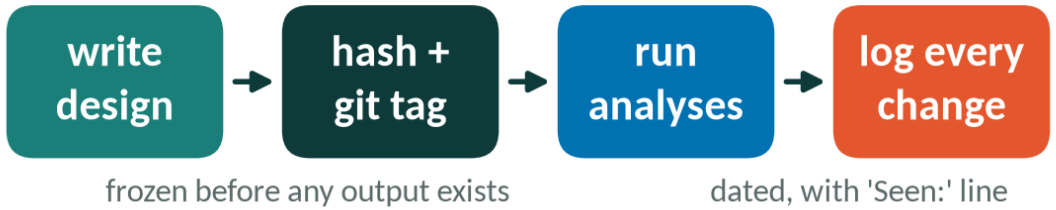
Right: geometric mean / pooled rate ratios from report Figure 3b. A tip looks 'novel' just for being a tip.

Phylogenetic regression in one picture



PGLS: regression whose errors follow the species tree (Pagel's λ) · 148 genomes · covariates: proteome size, disorder

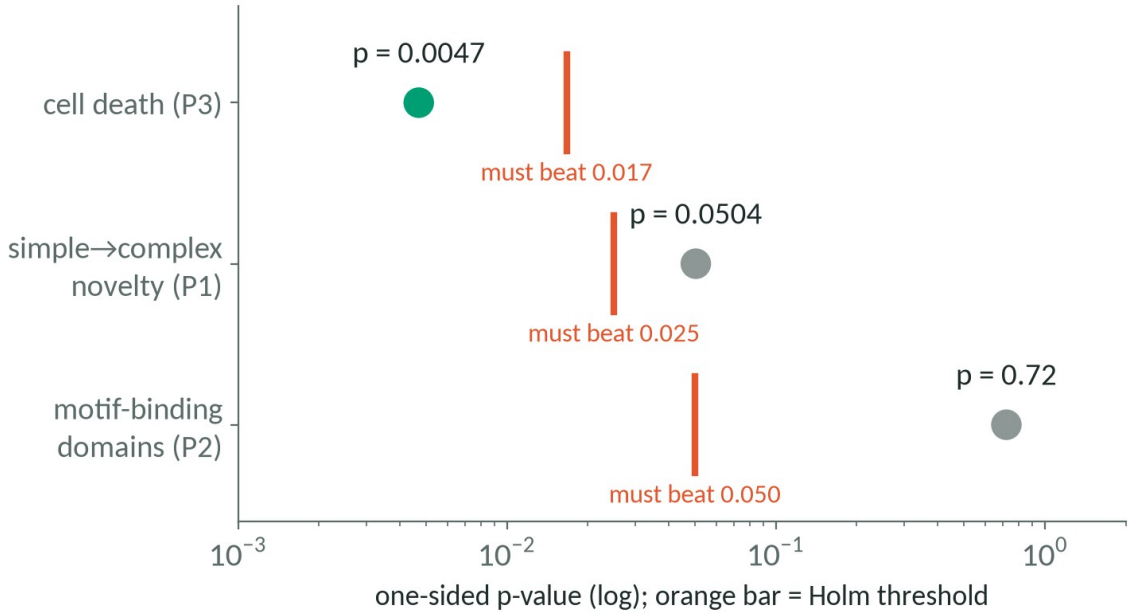
Pre-registration and multiple testing



prereg-v1
round 1 · 15 hashed inputs

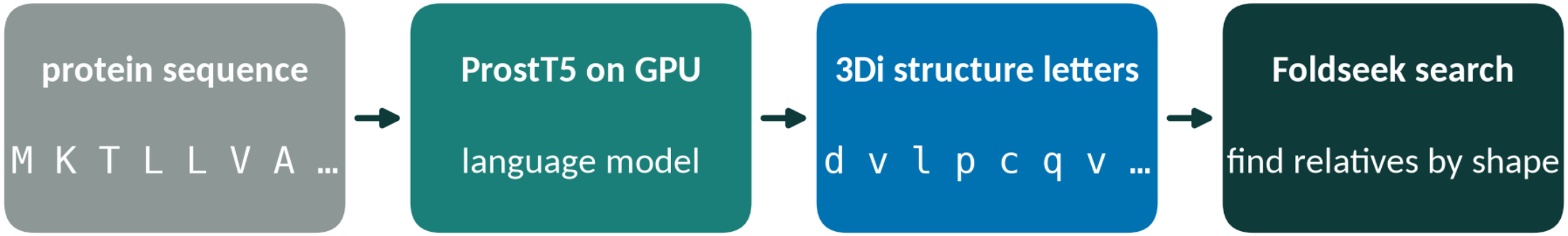
prereg-r2
round 2 design + family lists

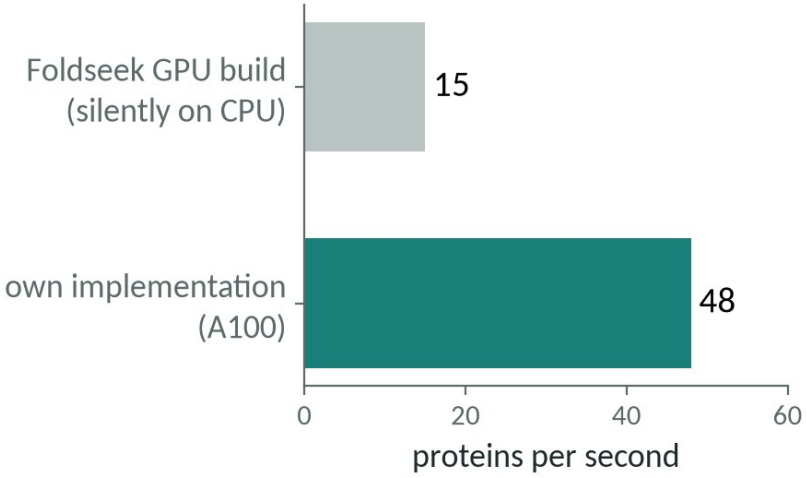
prereg-r2-p1
4 confirmatory events

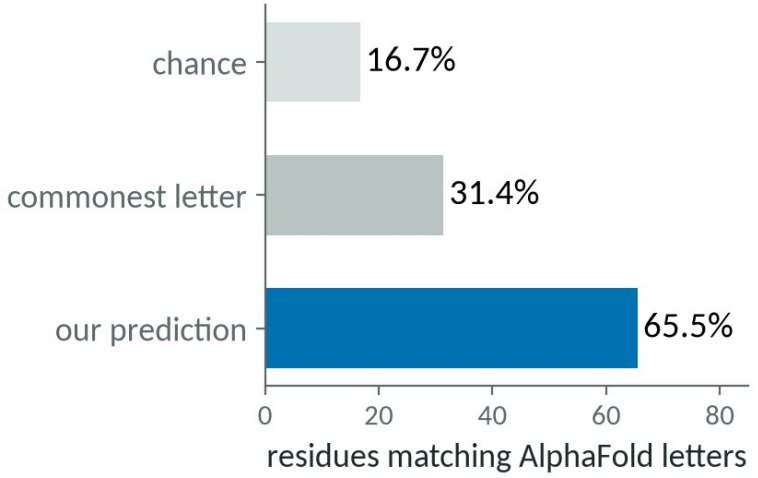


Holm: smallest p must beat α/3, the next α/2, …; stop at the first failure.

Structure from sequence, on one GPU

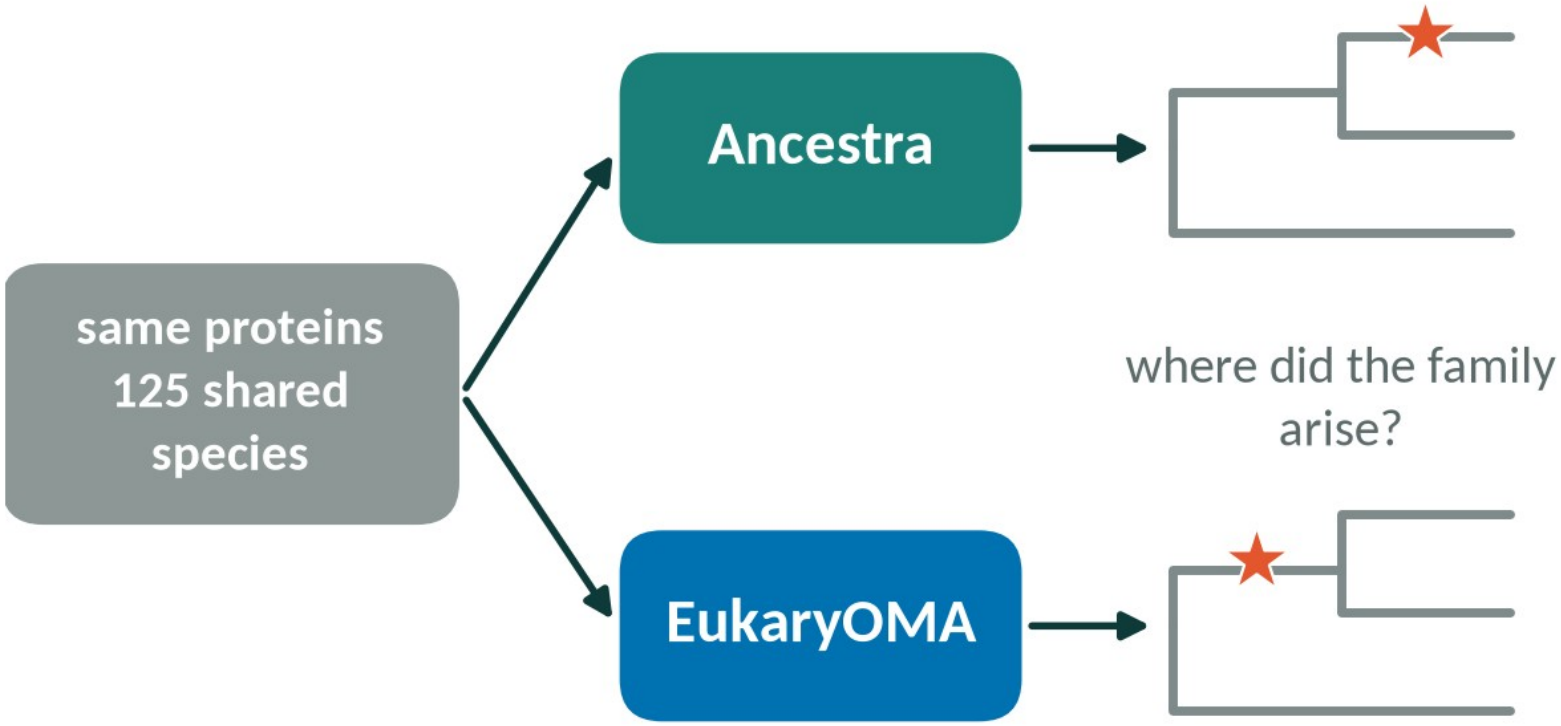






Lesson check the device is really used: the pilot caught a silent CPU fallback in the first hour

Cross-checking with a second method

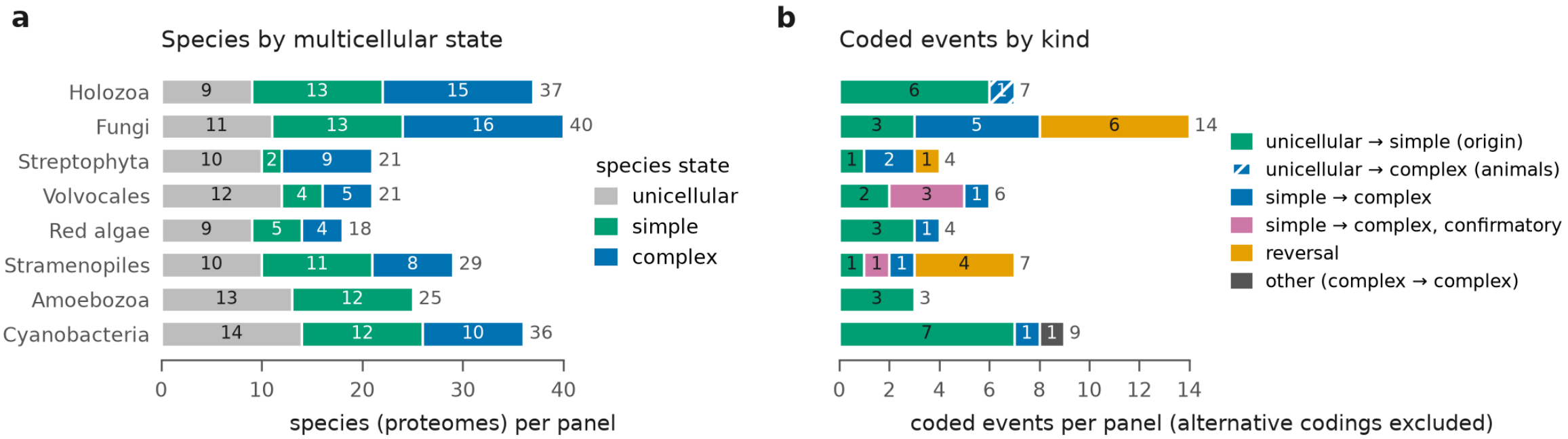


29%

same branch
(540 of 1,859 stem gains)

Same proteins, same tree, same parser: only the orthology method differs. Agreement measures consistency, not accuracy.

Round 2: eight focused reconstructions



18–40

proteomes per panel (227 in all)

15–45 min

per reconstruction (round 1: ~9–10 h)

5–9

control branches per panel: coarse tests

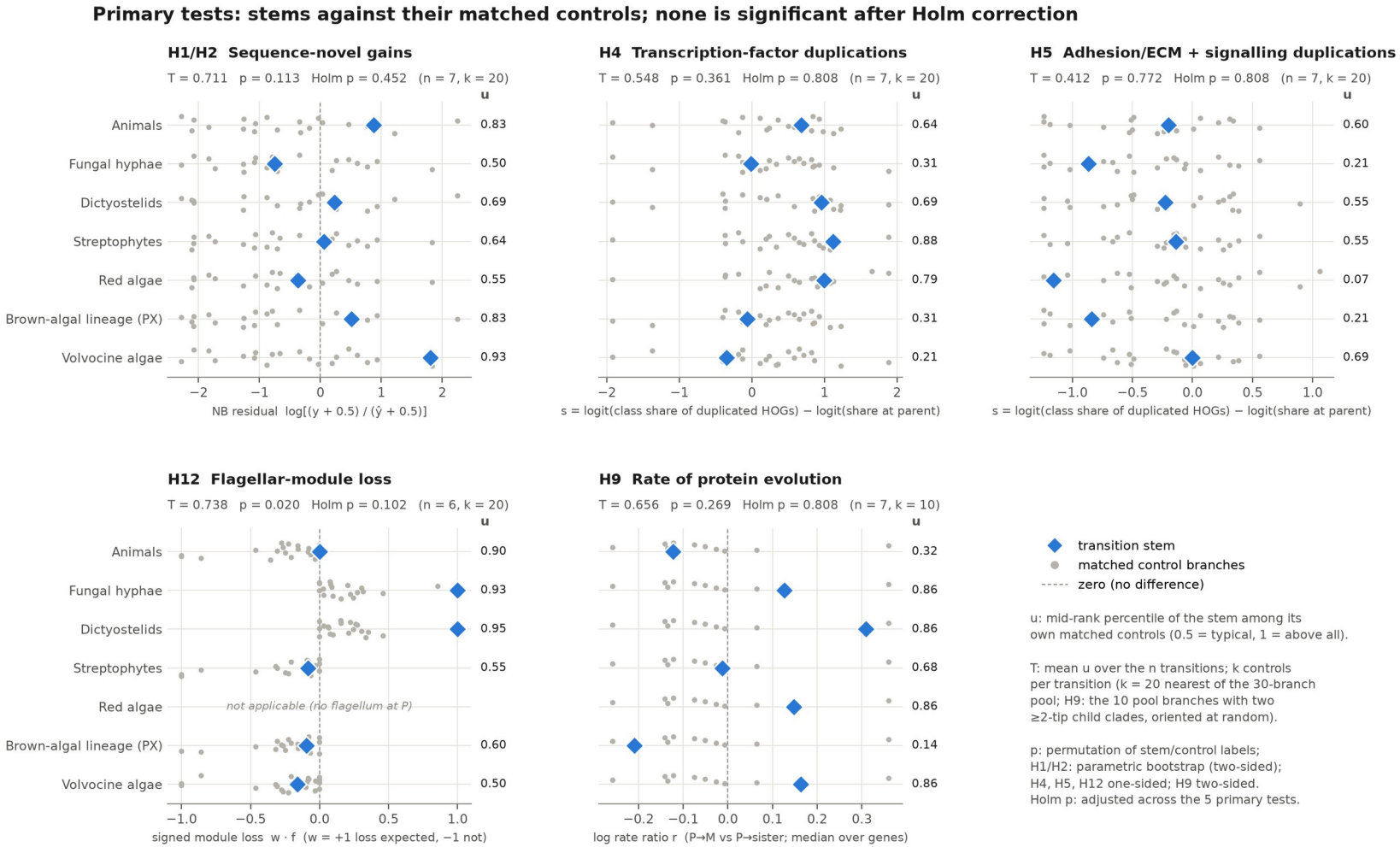
F

Main results

One message per slide



Round 1: no shared signature of seven origins



0 of 5

primary tests significant after Holm

1.10×

rate ratio [0.96–1.26]: no ≥1.5-fold shift

13–35%

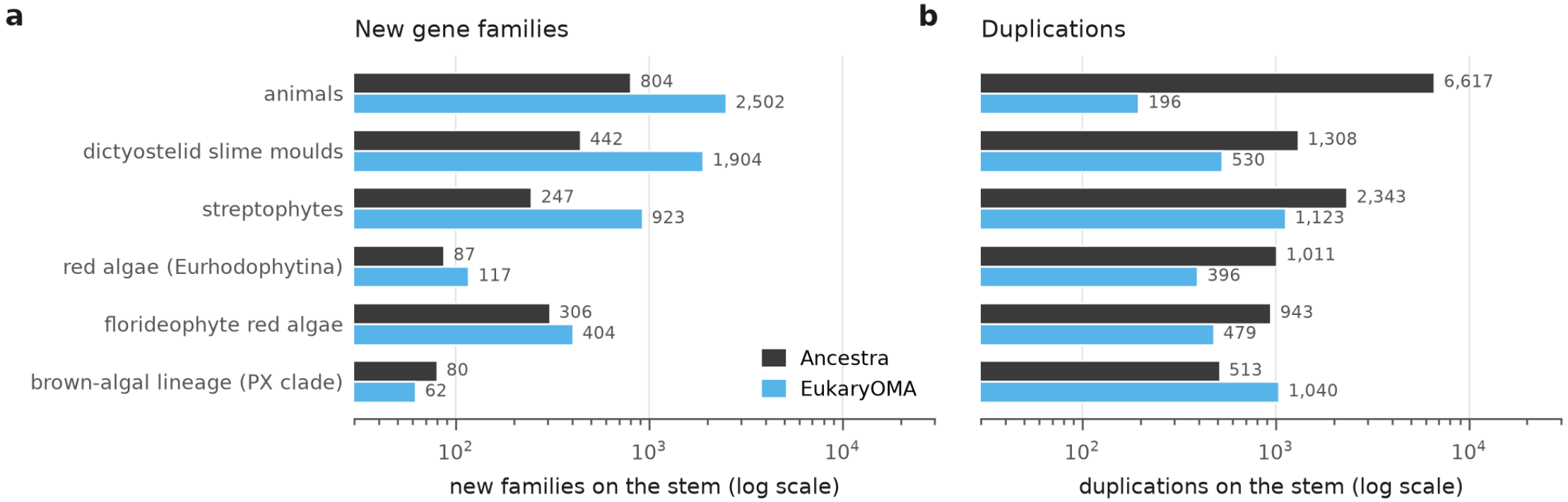
power of gene-content tests at 1.5-fold

Errata: the review caught three overstatements

Claim	Round 1 said		Corrected
Depth imbalance, stems vs controls	1.77 SD	→	2.06 SD
Positive controls that test Ancestra	'11 of 16'	→	6 of 9; deep-node 0 of 3
Rate-test p-value	seed-dependent 0.19–0.61	→	exact: p = 0.31
'Eukaryoma' database	'not found'	→	EukaryOMA: 1,328 genomes

None of these changes a round-1 conclusion.

Two orthology methods disagree on where genes arose



29%

of Ancestra stem gains on the same branch in EukaryOMA

9%

of EukaryOMA stem gains confirmed by Ancestra

4–8%

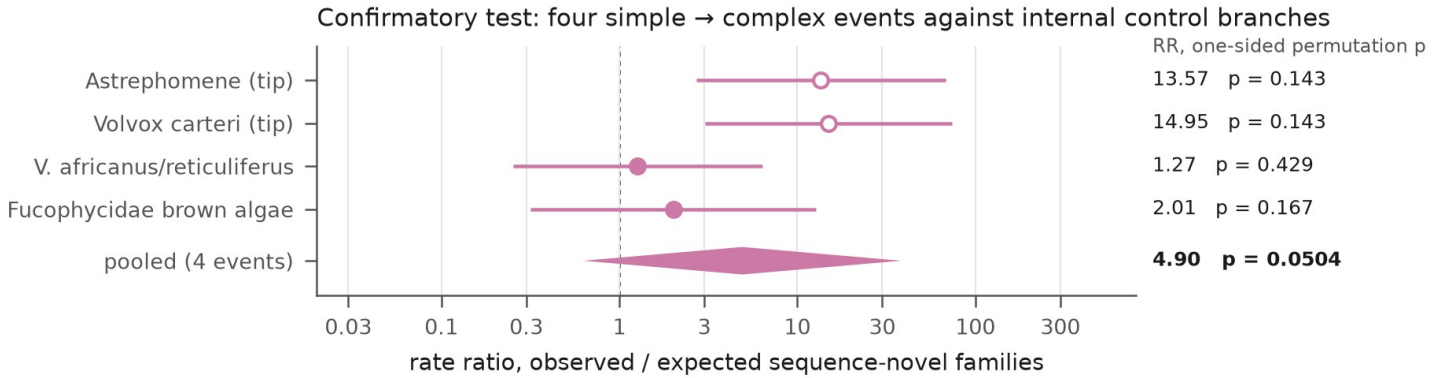
agreement on duplications

6,617 vs 196

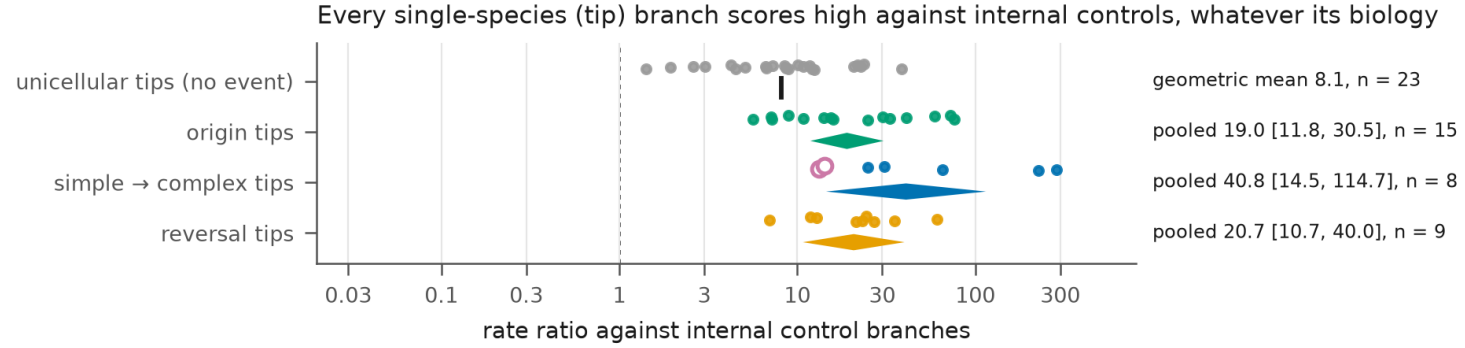
animal-stem duplications, Ancestra vs EukaryOMA

Simple to complex: no significant burst of new genes

a



b



p = 0.050

combined, Holm-adjusted 0.10

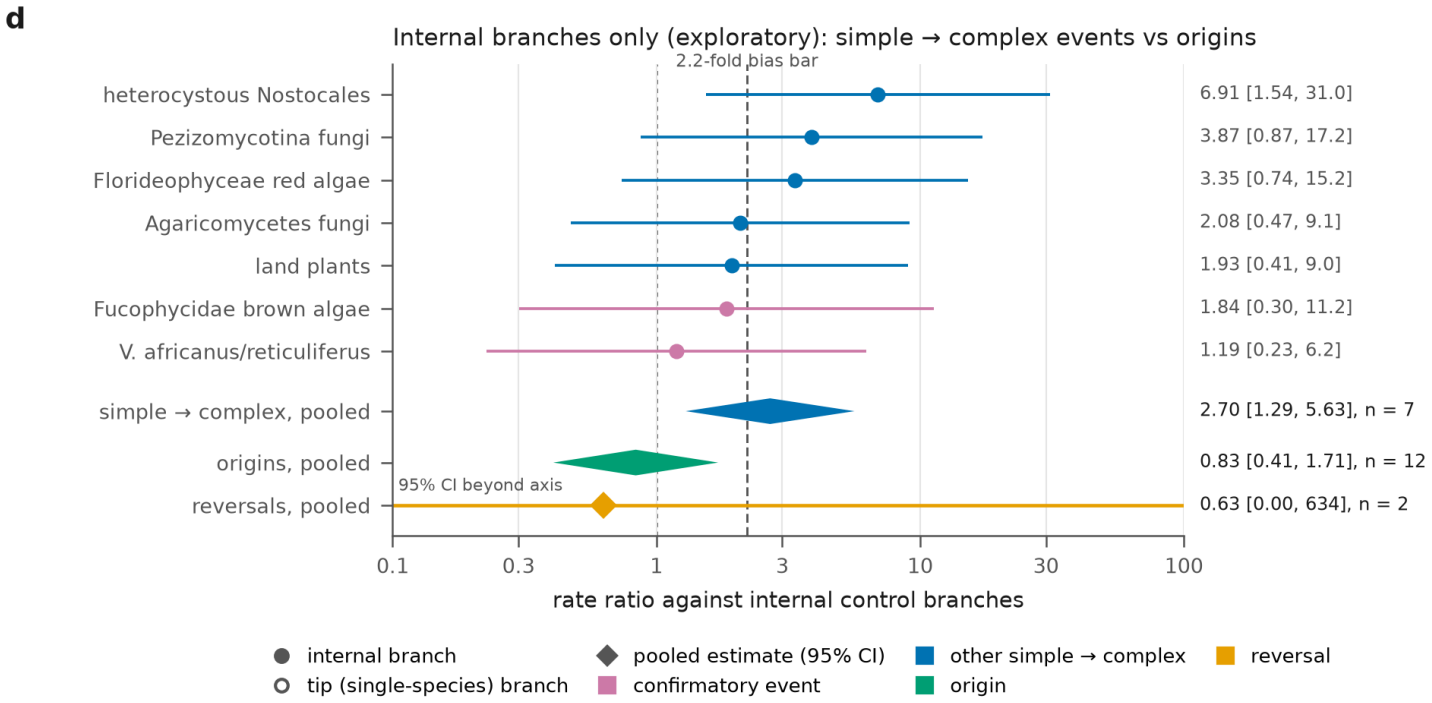
8.1×

unicellular tips with no event, vs internal controls

0.56–0.80

the tip events, compared tip with tip

Internal branches only: a weak lead



2.8×

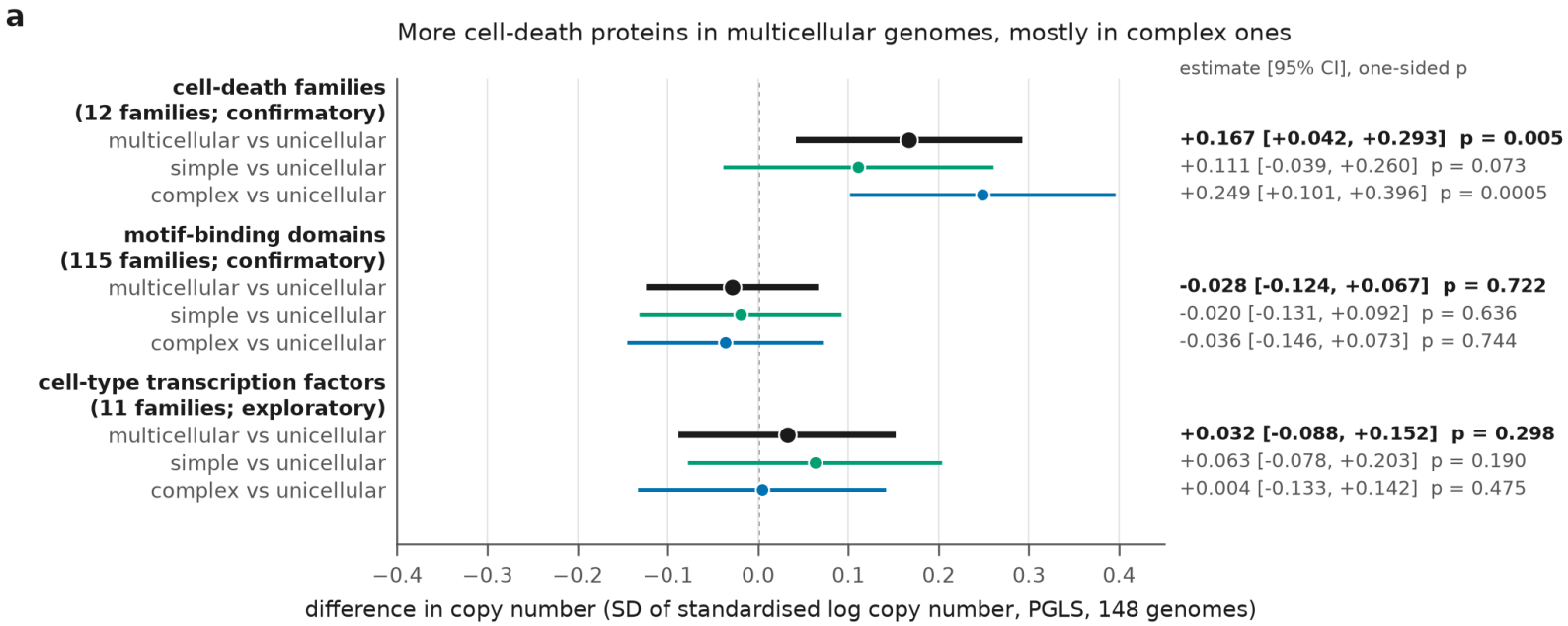
simple → complex, internal [1.34–5.89]

0.82×

origins of simple multicellularity

Lower bound below the 2.2× bias line: exploratory only

Cell death: the one confirmed signal



1.21×

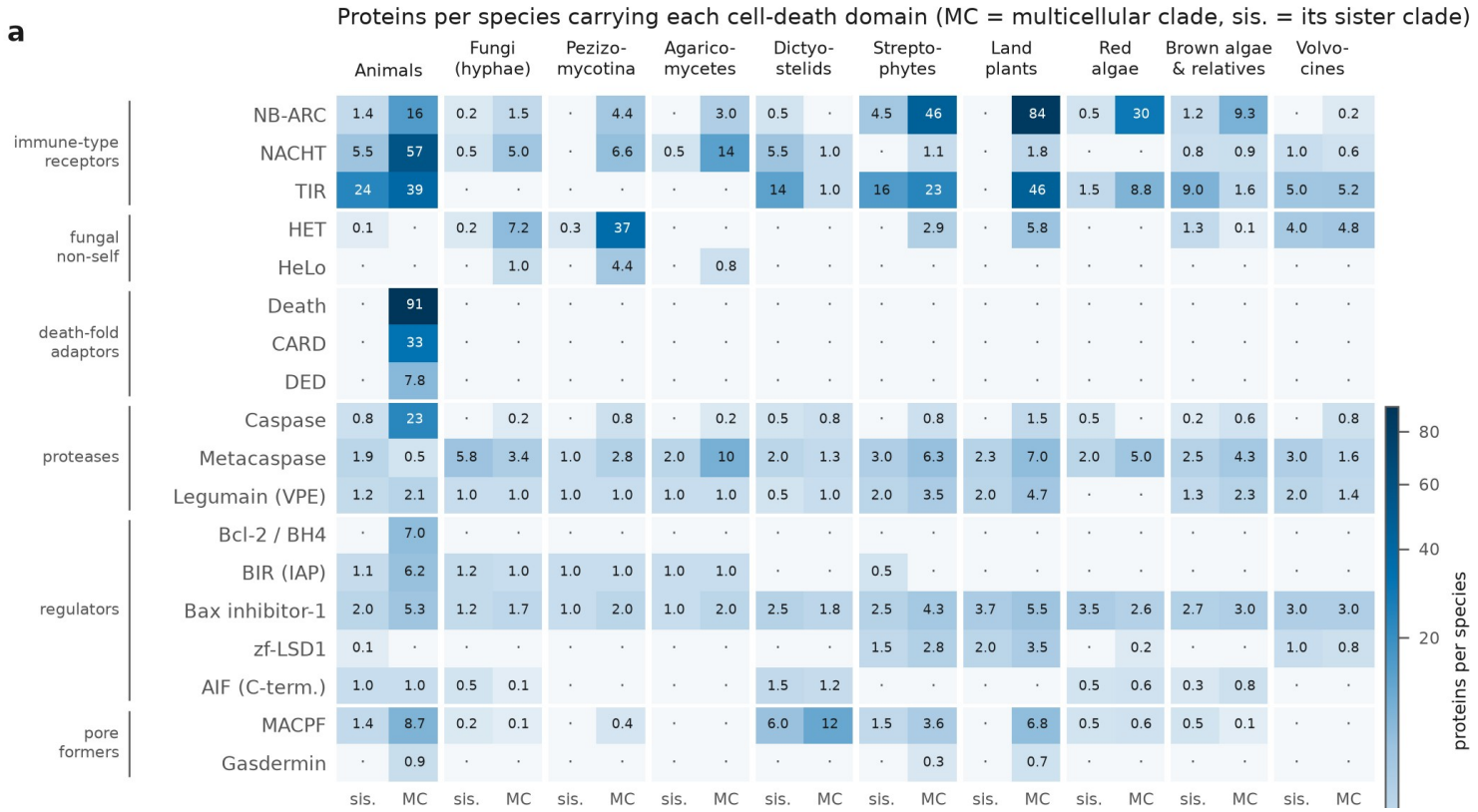
more cell-death proteins [1.04–1.42], Holm p = 0.014

+0.249 SD

complex vs unicellular (simple: +0.111)

Caveat: size-dependent. Without the 10 largest proteomes p = 0.025; as a share of the proteome p = 0.14. Read as 'more than expected for proteome size'.

The extra proteins are lineage-specific receptors



Land plants
NB-ARC–LRR immune receptors (+84 / species)

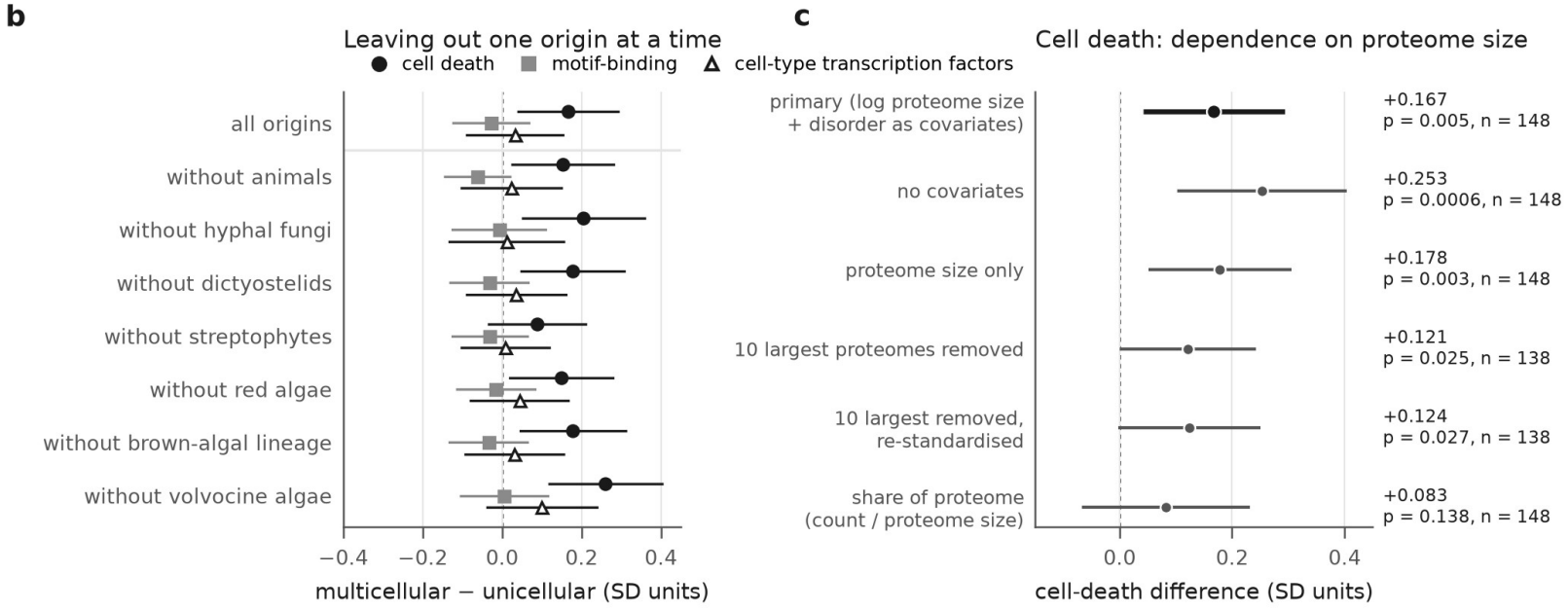
Red algae
NB-ARC–WD40 (+29)

Filamentous fungi
HET (+37), NACHT: non-self recognition

Animals
Death, CARD, DED adaptors; caspases

Dictyostelids, volvocines
no excess

Motif-binding domains and TF families: null



0.98×

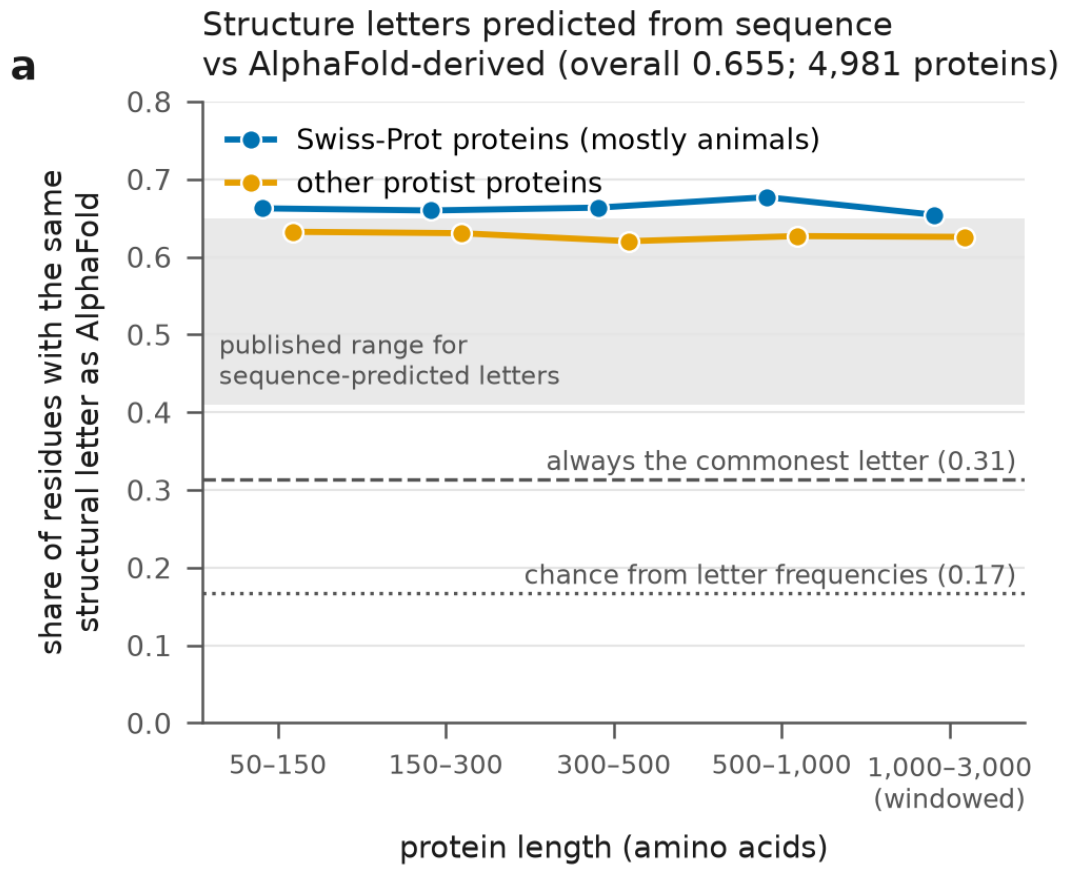
motif-binding domains, p = 0.72

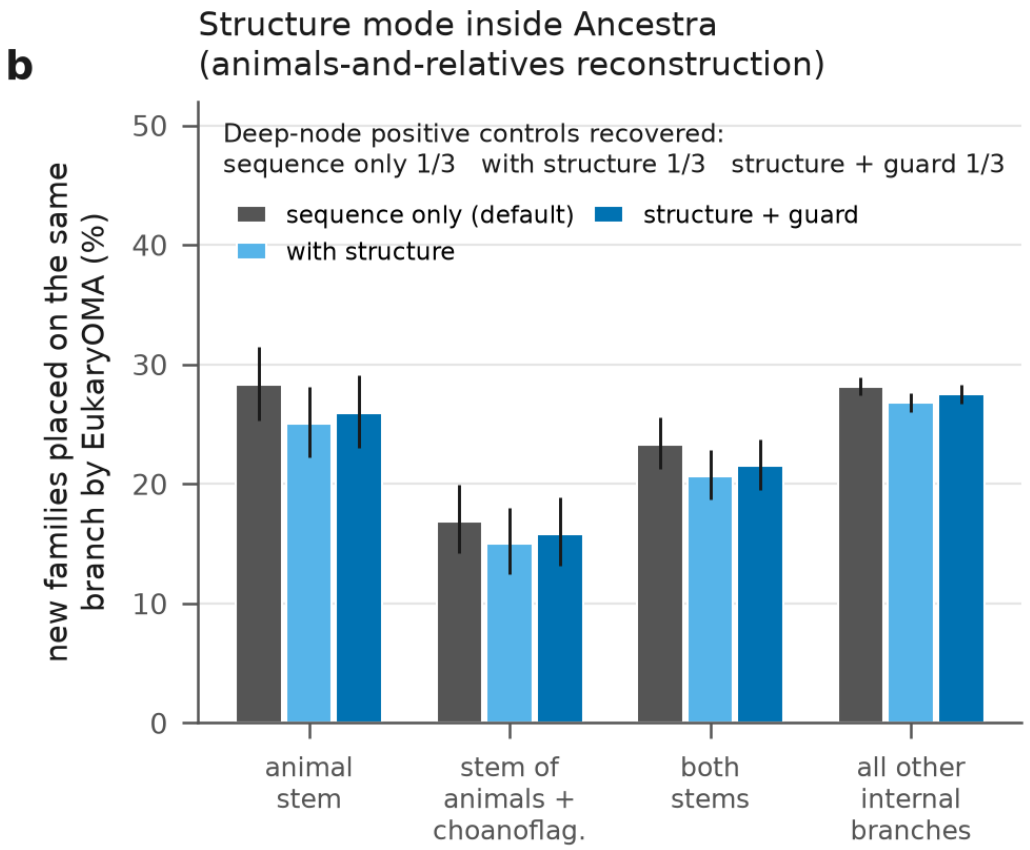
+0.03 SD

cell-type TF families, p = 0.30

Left: motif-binding estimates are negative in 6 of 7 leave-one-out runs; cell death stays positive in all 7.

Structure did not help Ancestra





65.5%

letters match AlphaFold (chance 16.7%)

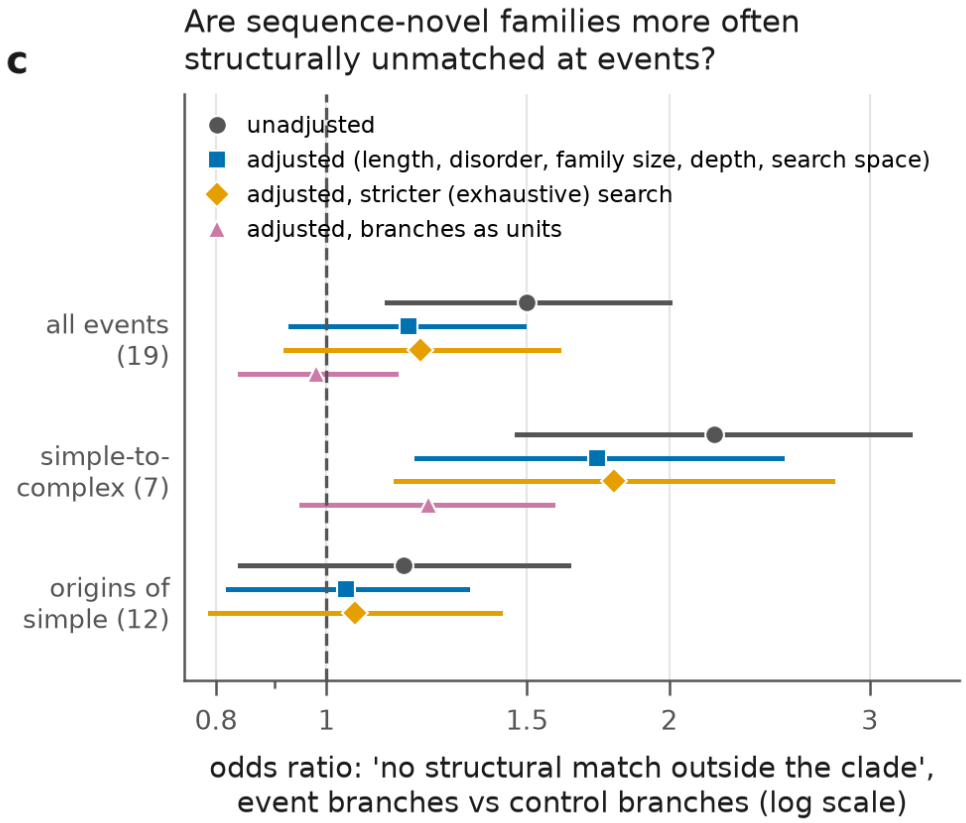
23.3% → 20.7%

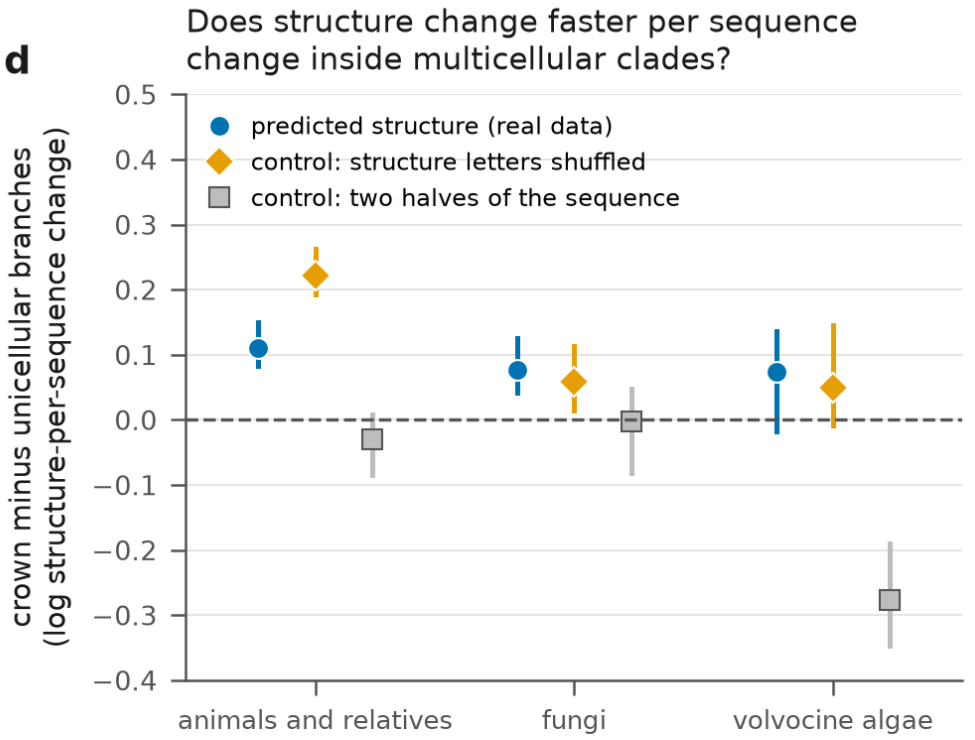
EukaryOMA agreement, default → structure

1 of 3

deep-node controls, in every run

Two structural signals did not survive controls





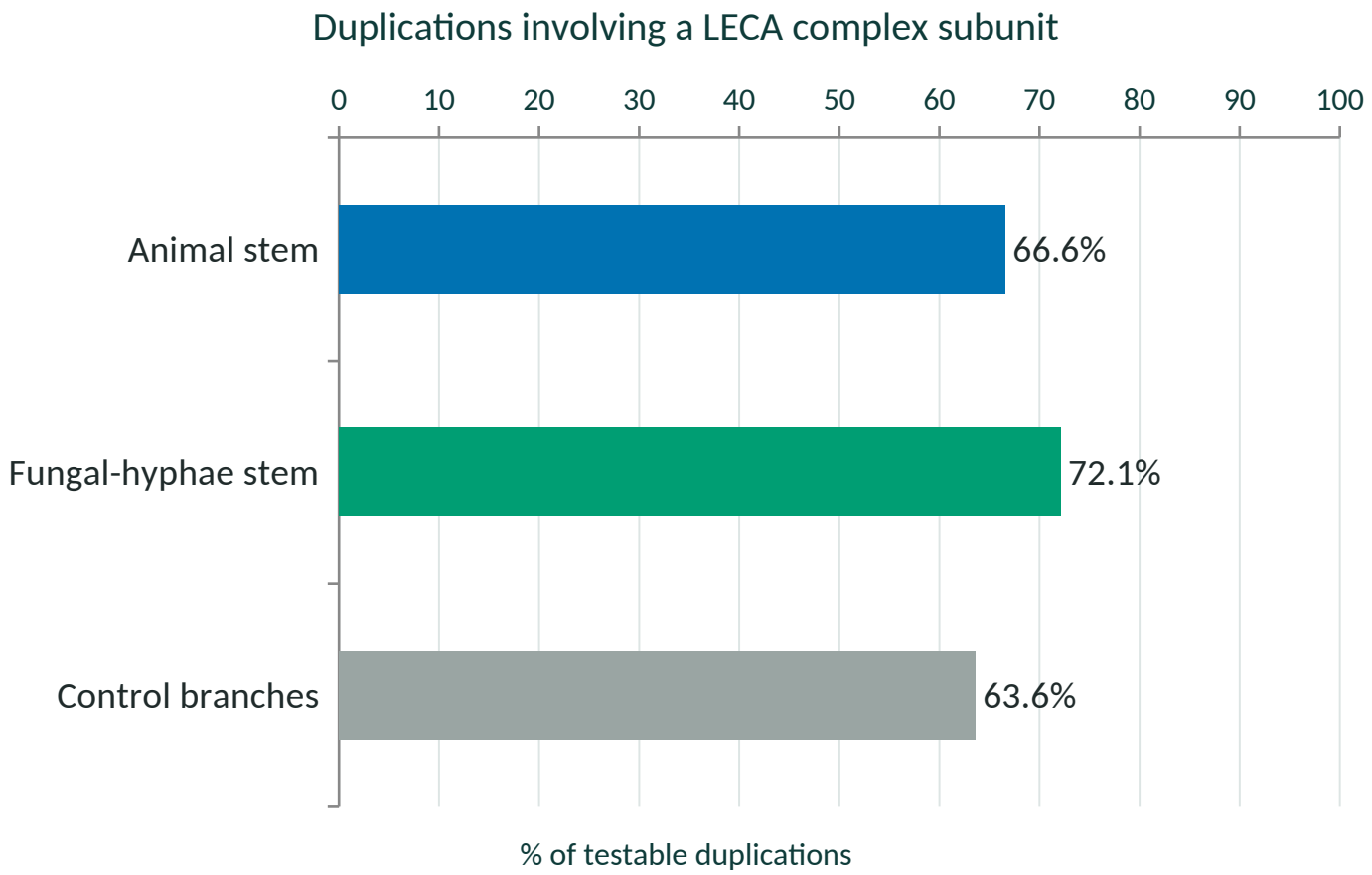
OR 1.44 → 1.18 [0.93–1.49]

after adjusting for length, disorder, depth

0.222 vs 0.110

shuffled control vs real data (animals)

Protein complexes add nothing detectable

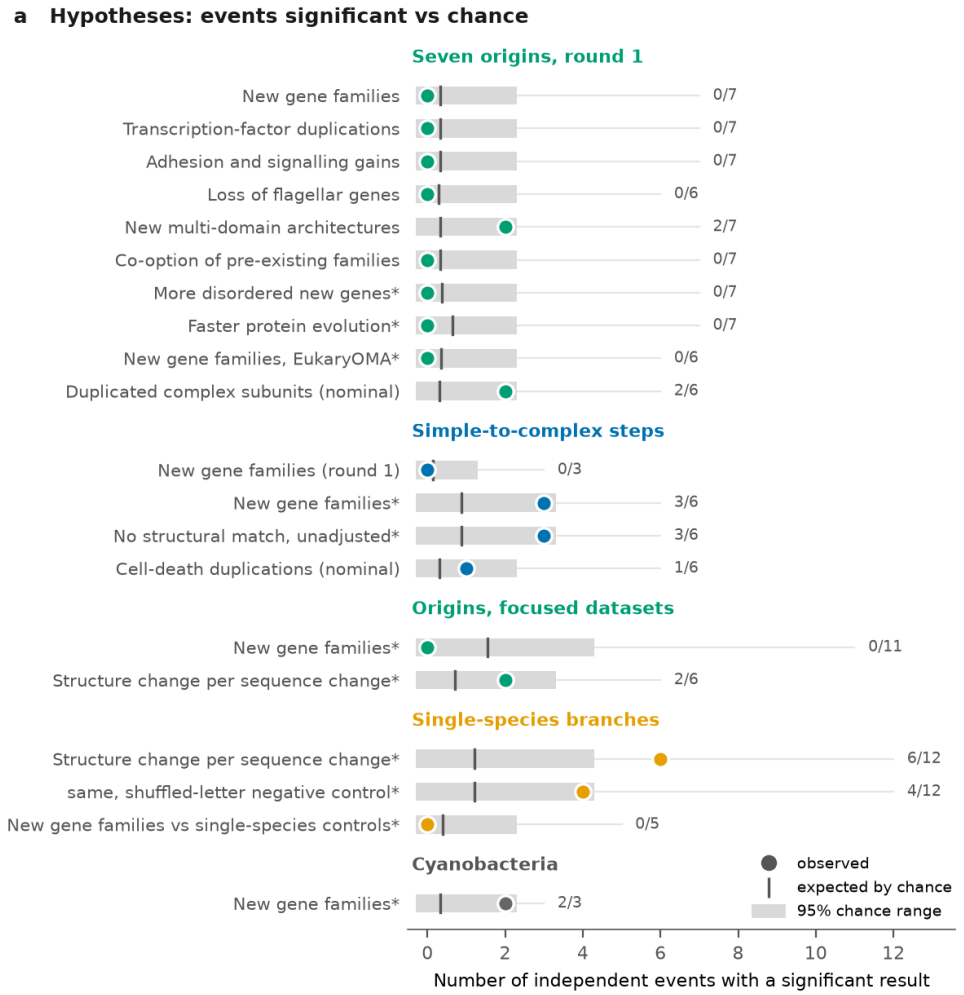


No complex enriched across stems after correction

- Map holds only ancient (LECA) complexes
- 13 species mappable
- Duplication counts are method-dependent

Data: Cox et al. 2026, LECA interactome from co-fractionation mass spectrometry.

Counting independent origins instead of averaging



0 / 7
new gene families

0 / 7
TF duplications

0 / 7
adhesion, signalling

0 / 6
flagellar loss

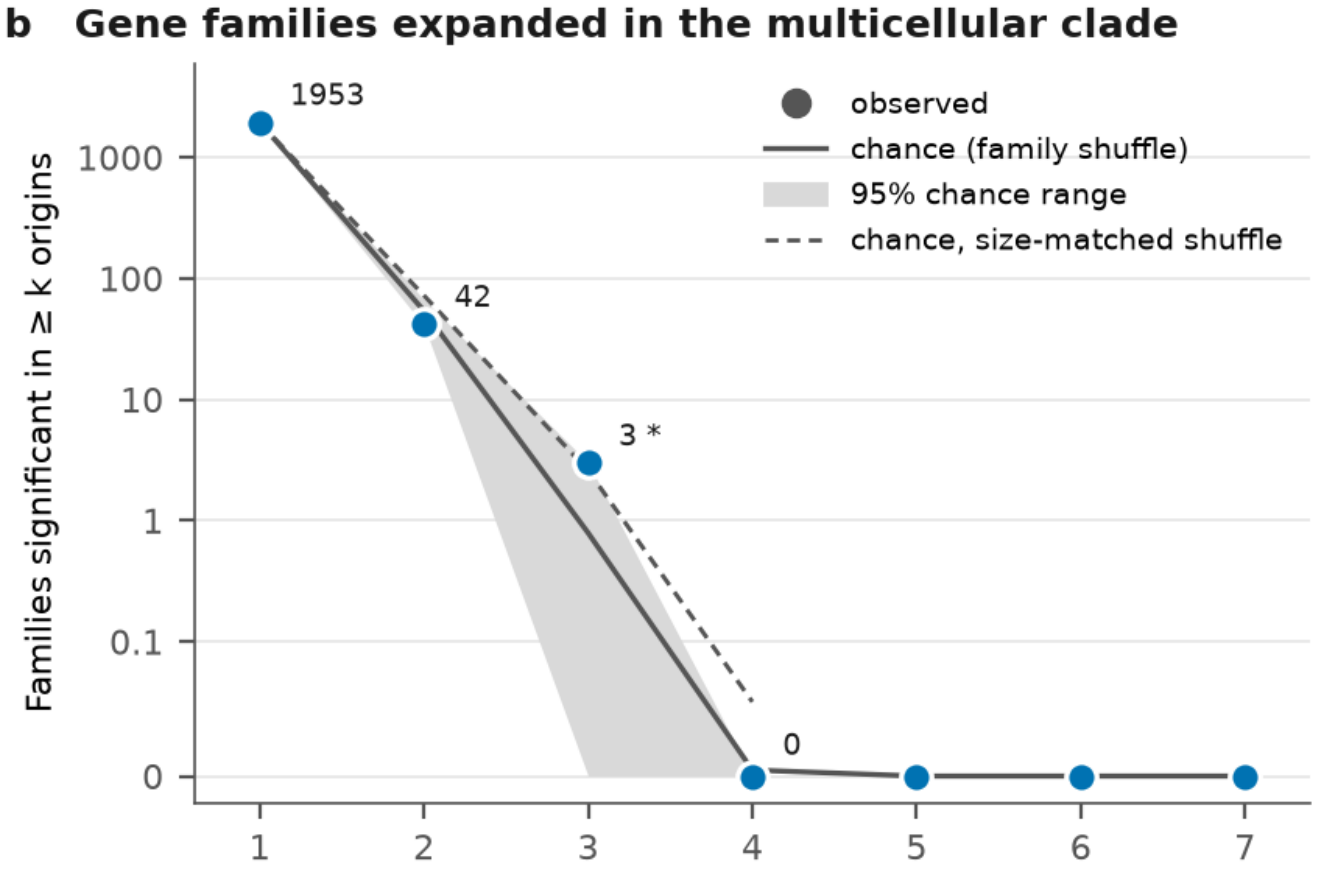
Each origin is null on its own
Counting does not rescue the nulls

- New domain architectures: 2/7, chance P = 0.04, Holm 0.65
- Simple→complex steps (focused panels): 3/6 top-ranked, P = 0.04, Holm 0.17

Cannot add per-event power; it avoids diluting signals. Each tier must be compared with chance.

Added after publication at the project lead's request (exploratory)

Recurrent families: contractions yes, expansions no



42 vs 54

expansions in ≥2 origins, observed vs expected

157 vs 90

contractions; genome-sisters only 29 vs 18

- Contractions inflated by transcriptome-sister artefacts
- 2 families pass correction: PDEase_I, Vint (q = 0.07)
- NB-ARC expanded in animals, red algae, land plants (nested step); primary origins only P = 0.09

Added after publication at the project lead's request (exploratory)

Take-home messages

1 **No shared signature** across origins; no ≥1.5-fold rate shift

2 **Cell death, ~1.2×** lineage-specific immune / self–non-self receptors

3 **Method matters** two orthology methods agree on 29% of stem gains

4 **Compare like with like** single-species branches always look 'novel'

5 **Structure: search, not signal** did not help Ancestra; signals failed controls

G

Every hypothesis, one by one

16 from the literature review, plus the round-2 tests

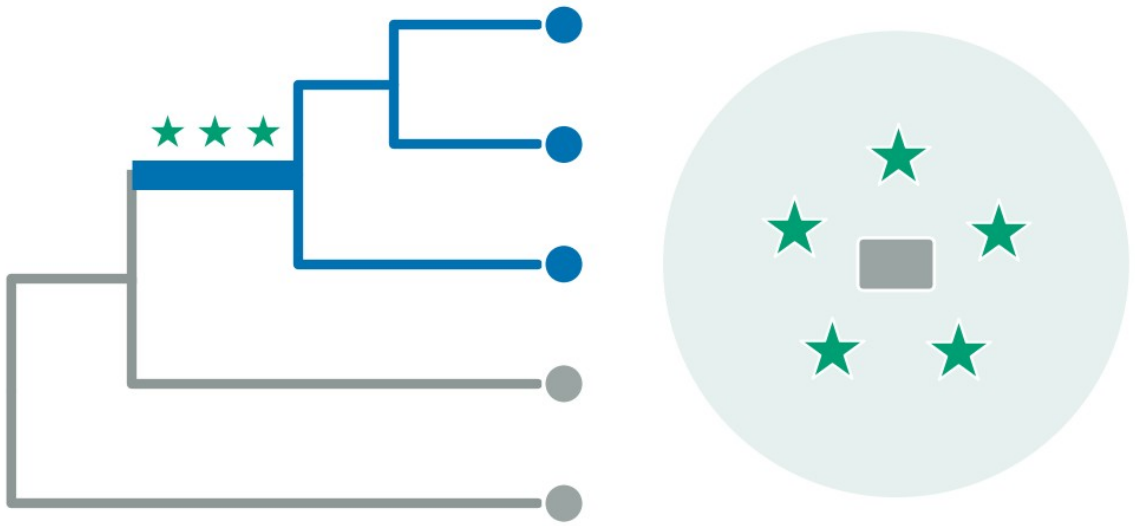


Scorecard of all hypotheses

H1–H2 co-option vs burst	**H3** pre-adaptation	**H4** TF expansion
H5 adhesion, signalling	**H6** receptor kinases	**H7–H8** aggregative vs clonal
H9 relaxed selection	**H10** disorder	**H11** domain shuffling
H12 loss of unicellular genes	**H13** cell-cycle co-option	**H14–H15** HGT, site convergence
H16 regulation, not coding	**R2-P1** simple → complex novelty	**R2-P2** motif-binding domains
R2-P3 cell death	**R2** structural novelty	**R2** protein complexes

supported · not detected / n.s. · not supported / null · exploratory lead · not tested

H1–H2 · Co-option or a burst of new genes?



Predicts Burst: more new gene families on transition stems than on controls. Co-option: no excess; the genes were already there.

Bursts reported at animal, land-plant and brown-algal origins; choanoflagellates already carry cadherins and kinases.

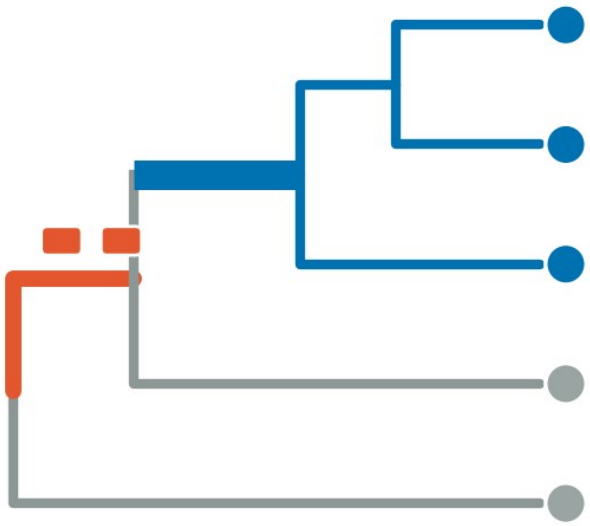
Gain counts inflate 2.1–2.6×; denser outgroup sampling moves 'animal-specific' families earlier.

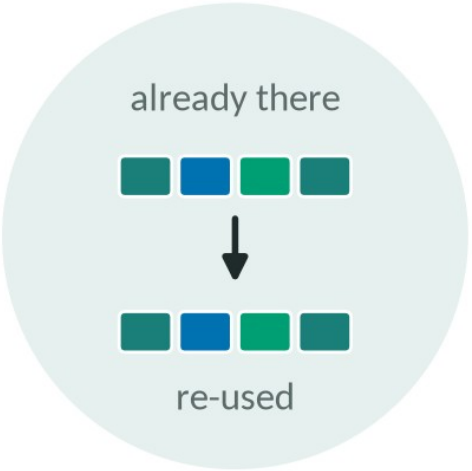
Literature Paps & Holland 2018; Bowles et al. 2020; Denoeud et al. 2024; King et al. 2008; Sebé-Pedrós et al. 2010; Prochnik et al. 2010; Kiss et al. 2019; Richter et al. 2018

NOT DETECTED (LOW POWER)

pooled 1.67× [0.64–4.34], p = 0.11 (Holm 0.45); 80–100% of expanded families present before

H3 · Pre-adaptation before the transition?





Predicts Ancestors of lineages that later turned multicellular already had more adhesion, TF and signalling genes than sisters that never did.

'Predisposition' reported for complex fungi; gradual assembly of the toolkit in holozoans.

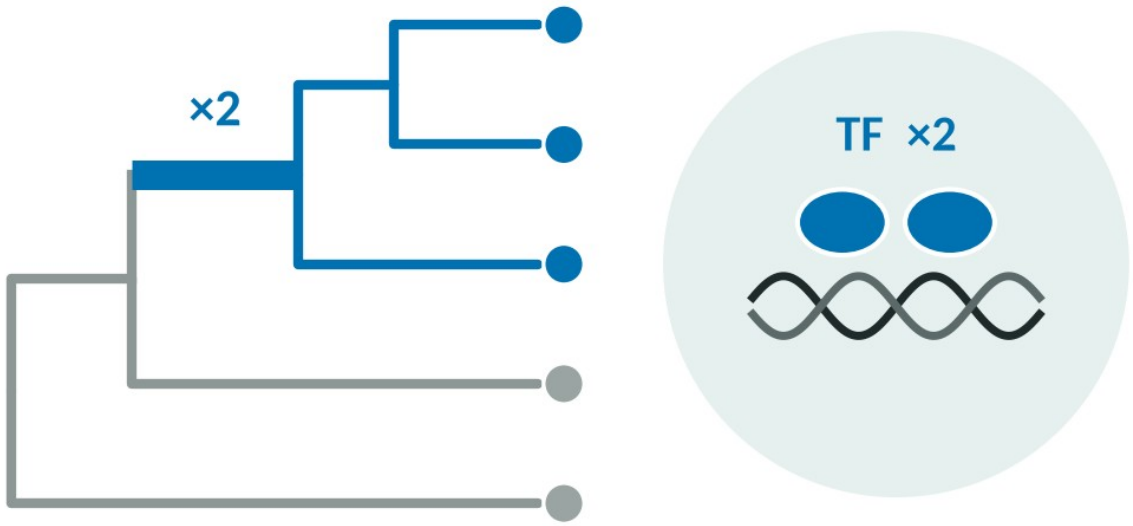
Needs many transitioning and non-transitioning sister pairs with good genomes.

Literature Ocaña-Pallarès et al. 2022; Grau-Bové et al. 2017; Merényi et al. 2020

NOT TESTED

not run in either round

H4 · Do transcription-factor families expand?



Predicts More TF duplications on transition stems than on controls, strongest in complex lineages.

TF repertoires track cell-type number; expansions reappear within origins (T-box, Runt in animals; TEA in fungi; SBP in plants).

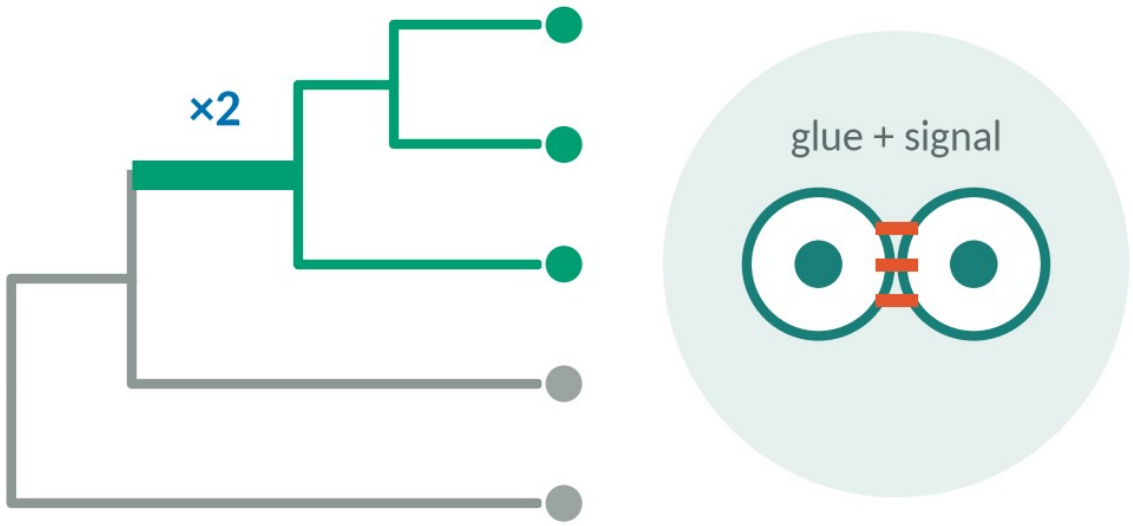
Much of the TF toolkit predates multicellular lineages.

Literature de Mendoza et al. 2013; Larroux et al. 2008; Lang et al. 2010; Catarino et al. 2016; Knoll 2011; Nagy et al. 2014

NOT DETECTED (LOW POWER)

R1: odds ratio 1.16 [0.62–2.19], p = 0.36 · R2 TF families: +0.03 SD, p = 0.30

H5 · Convergent adhesion and signalling expansion?



Predicts The same adhesion/ECM and signalling domains expand independently in two or more lineages.

Every multicellular body needs cell adhesion and communication.

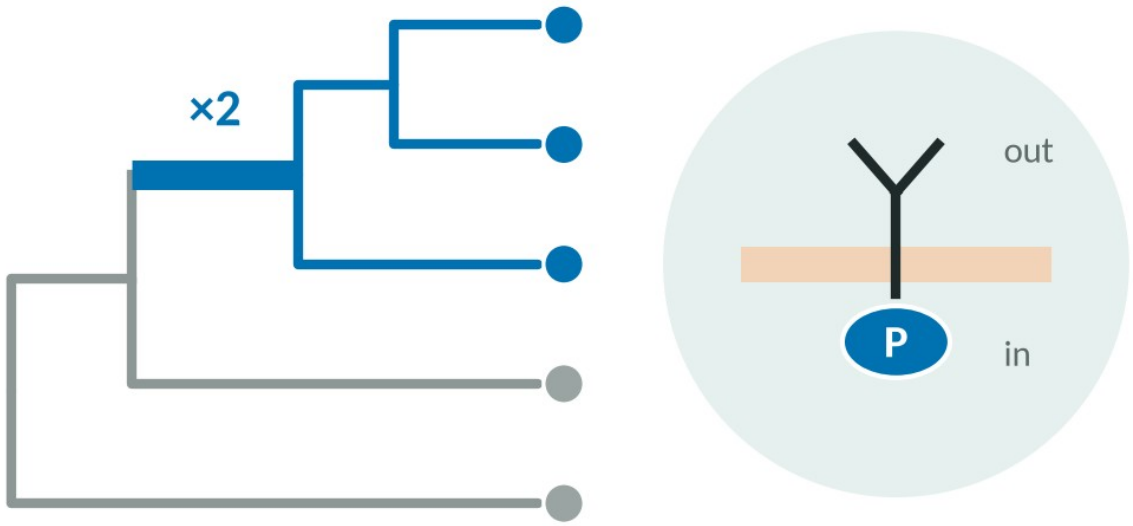
Classical cadherins are new domain combinations inside old families, invisible to family-level dating.

Literature Knoll 2011; Abedin & King 2010; Pereira Lobo et al. 2025; Rokas 2008a

NOT DETECTED (LOW POWER)

odds ratio 0.83 [0.55–1.27], p = 0.77; Pfam-wide expansions not above chance

H6 · Receptor kinases in complex lineages?



Predicts Receptor-kinase families duplicate on brown-algal, plant and animal stems, but not on fungal hyphae.

Receptor kinases expanded in Ectocarpus and Arabidopsis; R2: one new kinase family at 5 of 7 complex events.

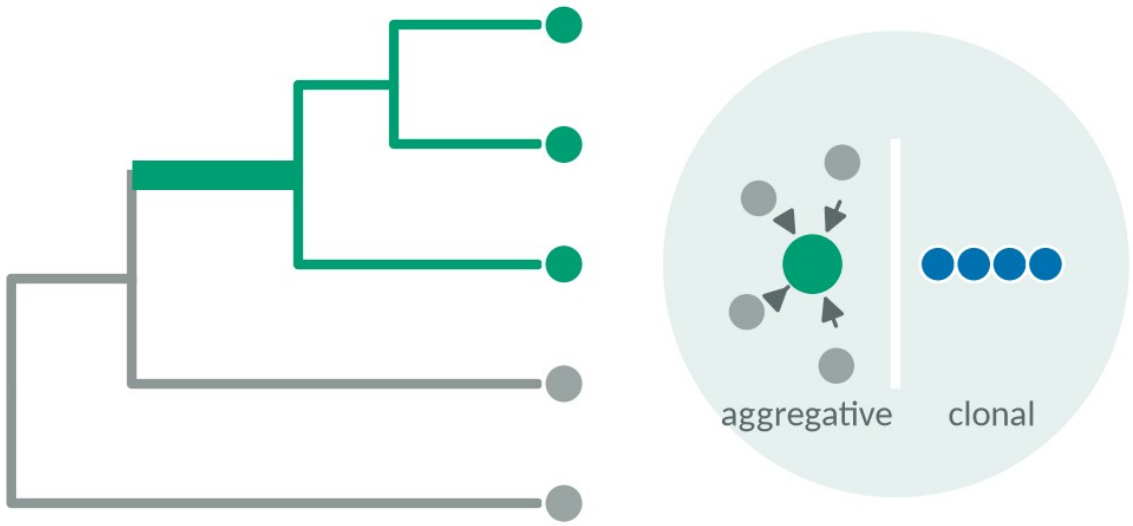
Receptor-kinase classification not run; kinase recurrence is only suggestive (q = 0.06).

Literature Cock et al. 2010; Shiu & Bleecker 2001; Manning et al. 2008; Kiss et al. 2019

PARTLY TESTED

fungal 'no expansion' control passed; kinase recurrence p = 0.023, q = 0.06

H7–H8 · Aggregative versus clonal



Predicts H7: aggregative origins gain secreted and membrane signals. H8: clonal origins show more gene turnover.

Dictyostelid development relies on extracellular signals; clonal development allows more cell types.

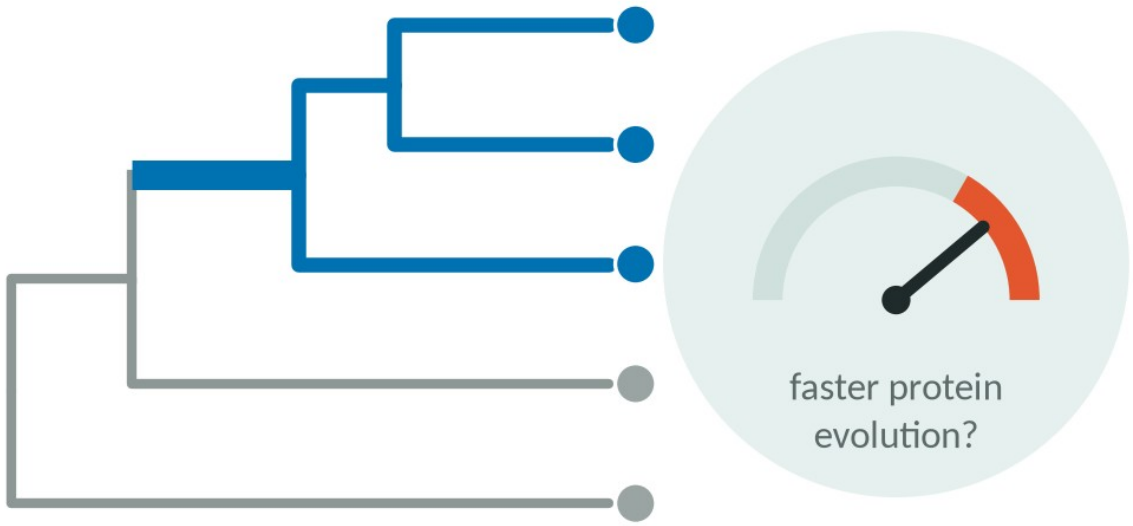
Few aggregative lineages with genomes; most are single species, so the stem is also a tip.

Literature Glöckner et al. 2016; Grosberg & Strathmann 2007; Fisher et al. 2013; Márquez-Zacarías et al. 2021

NOT TESTED

one aggregative primary origin (dictyostelids); not run

H9 · Relaxed selection after the transition?



Predicts Smaller populations weaken selection: faster protein evolution and larger proteomes after eukaryotic transitions.

Drift theory of genome complexity, recently extended to multicellularity.

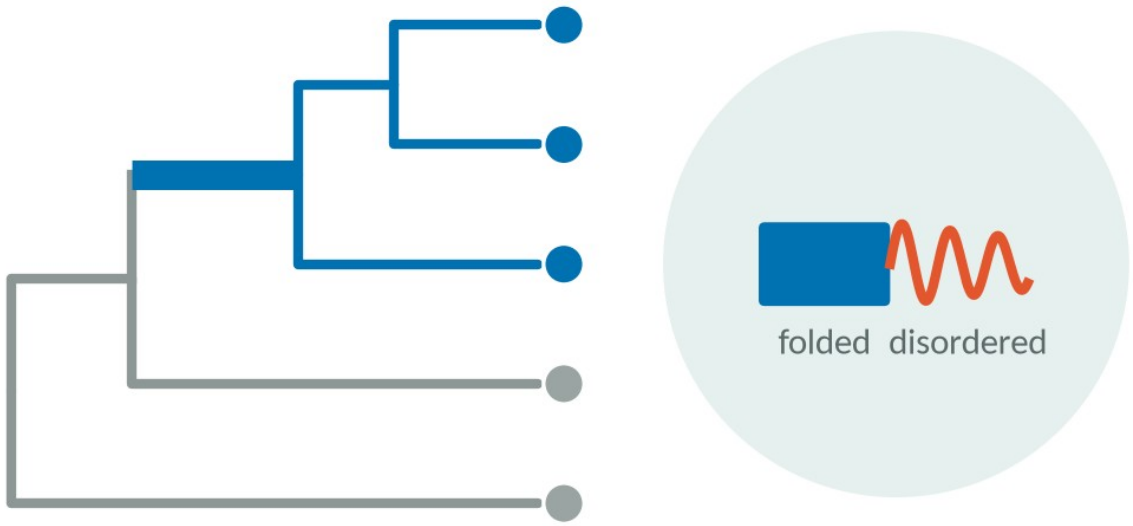
Rates differ by lineage (4 stems faster, 3 slower); R2 structure-per-sequence test failed its control.

Literature Lynch & Conery 2003; Lynch 2007; Bingham & Ratcliff 2024; Pentz et al. 2023

NOT SUPPORTED

pooled rate ratio 1.10 [0.96–1.26]: no ≥1.5-fold shift; exact p = 0.31

H10 · More disorder with complexity?



Predicts Disorder in transcription factors rises with cell-type number and on transition stems.

Disorder and alternative splicing scale with organismal complexity in earlier surveys.

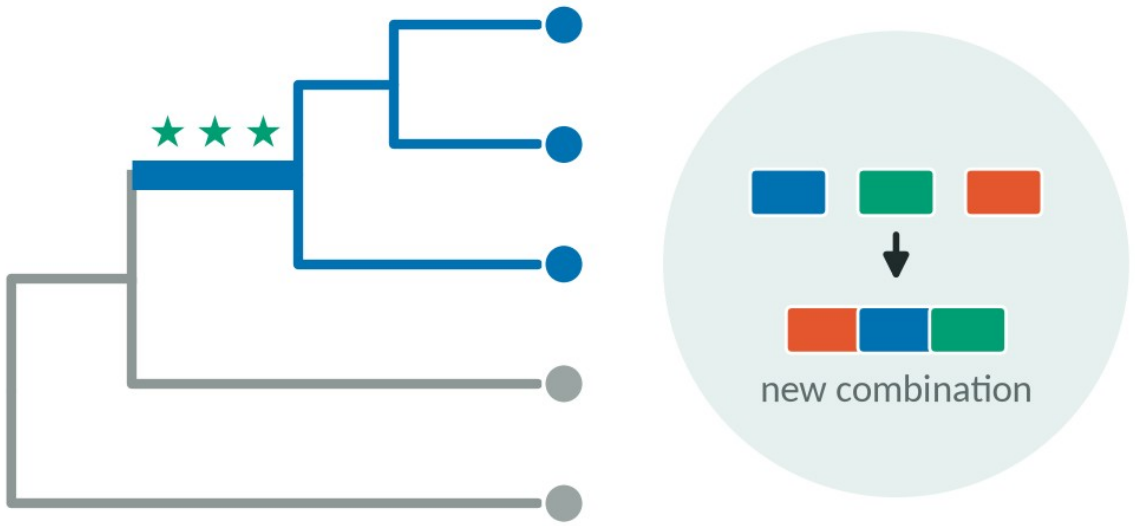
Mixed directions: disorder up in 4 of 7 transitions, down in volvocines and animals.

Literature Schad et al. 2011; Yruela et al. 2017; Chen et al. 2014; Dunker et al. 2015

NOT DETECTED (LOW POWER)

percentiles vs controls 0.34–0.73; regression on cell-type number not run

H11 · New domain combinations?



Predicts More new multi-domain architectures on complex-lineage stems than on controls.

Domain shuffling builds new proteins; animal stem: 1,131 new architectures vs control median 34.

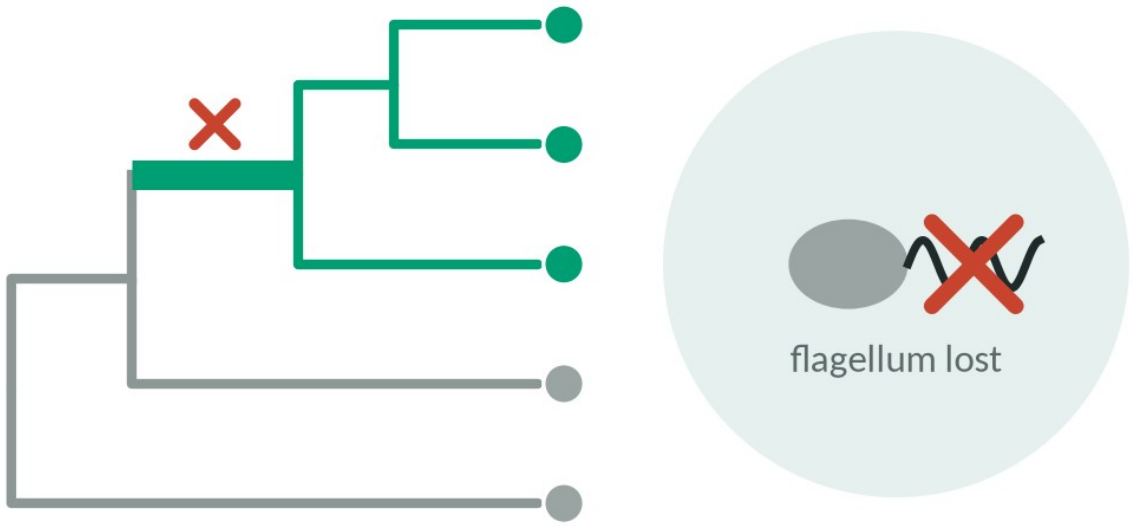
Fails a completeness filter (BUSCO ≥ 80%: p = 0.16); exploratory only.

Literature Ekman et al. 2007; Denoeud et al. 2024; Ocaña-Pallarès et al. 2022

EXPLORATORY LEAD

p = 0.005, survives leave-one-out, not robust to data quality

H12 · Shedding unicellular functions?



Predicts Flagellar and phagotrophy genes are lost at transitions; reversals lose multicellularity genes.

Fungi and dictyostelids lost all four flagellar modules, as expected ('shedding of protist genes').

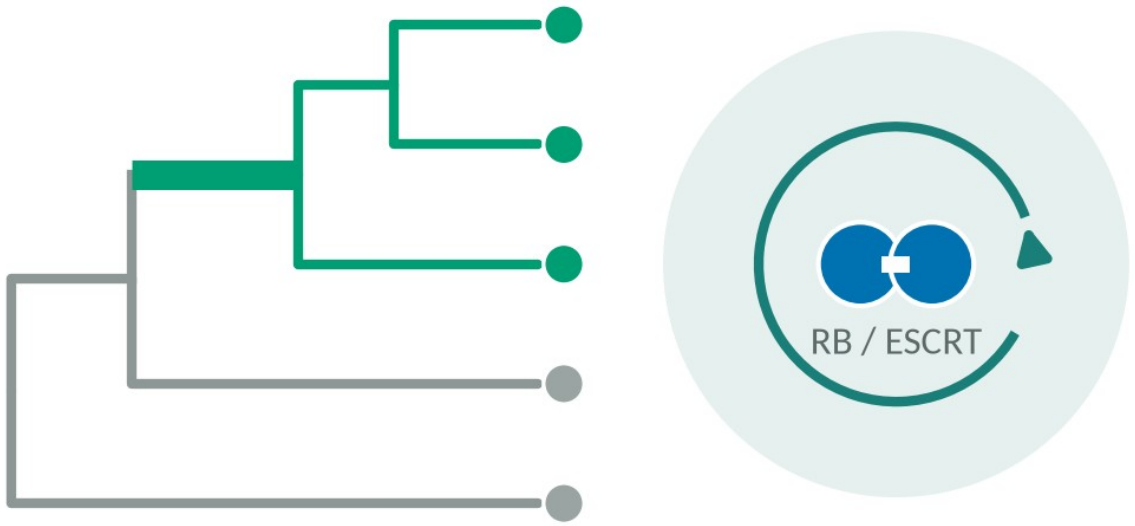
Only two loss-expected stems (power ≤ 0.33); reversals do not preferentially shed transition genes.

NOT DETECTED (LOW POWER)

p = 0.020, Holm-adjusted 0.10; reversal odds 0.86–1.39

Literature Merényi et al. 2023; Yang et al. 2026; Nguyen et al. 2017

H13 · Is cell-cycle control co-opted?



Predicts RB/E2F, ubiquitin–proteasome and ESCRT/septin genes duplicate or speed up on clonal stems only.

Gonium co-opted the retinoblastoma pathway; incomplete cell division links cells by bridges.

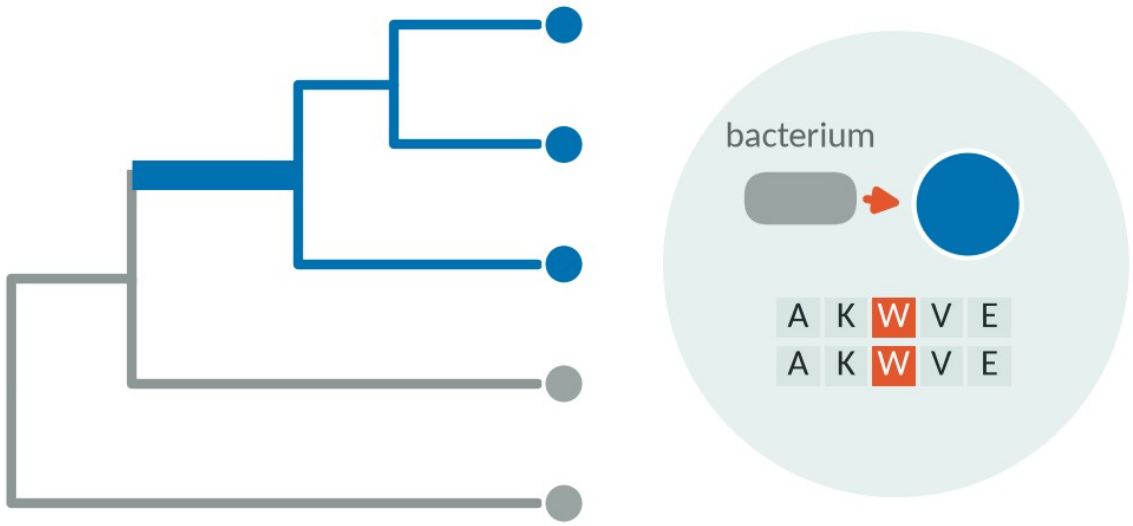
Duplication and rate contrasts were not run; only one co-option control.

Literature Hanschen et al. 2016; Featherston et al. 2018; Chaigne & Brunet 2022

NOT TESTED

RB pathway already present in the unicellular volvocine ancestor (control recovered)

H14–H15 · Gene transfer and site convergence



Predicts H14: non-animal origins gain more genes by horizontal transfer. H15: the same amino-acid changes recur at independent origins.

Gene transfer is common in fungi, protists and brown algae; strong molecular convergence in complex fungi.

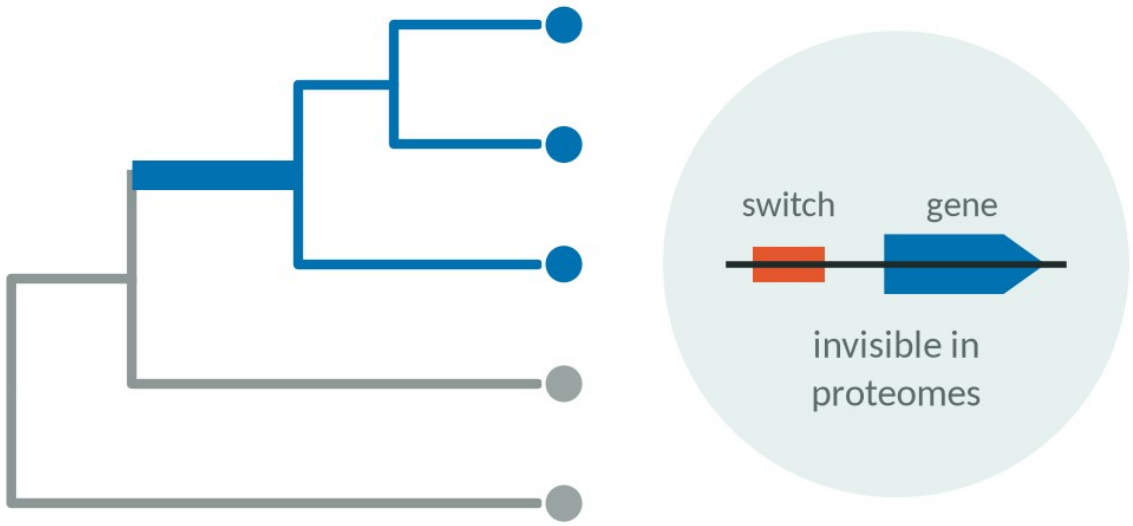
Both need per-family gene trees; cut for time at the design stage.

Literature Ocaña-Pallarès et al. 2022; Denoeud et al. 2024; Sheikh et al. 2024; Glöckner et al. 2016; Merényi et al. 2020; Iizuka et al. 2026

NOT TESTED

cut for time (DESIGN.md §4.2)

H16 · Regulation rather than new proteins?



Predicts Multicellularity mainly rewires gene regulation; protein-coding content changes little.

Dynamic regulatory genome in Capsaspora; our many nulls are compatible with it.

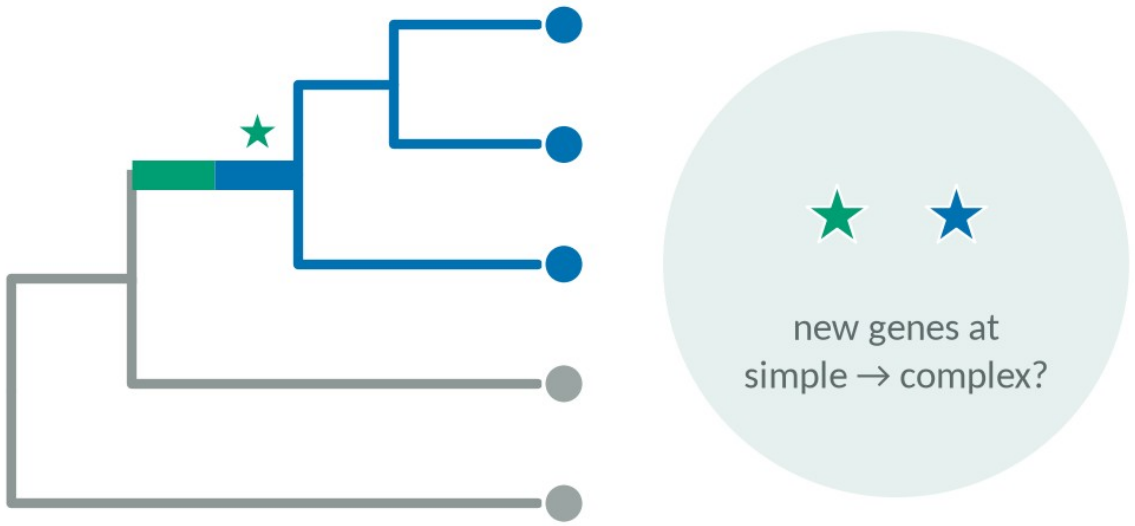
Regulatory change is invisible in proteomes; compatibility is not evidence.

NOT TESTABLE HERE

outside the scope of protein-only data

Literature Sebé-Pedrós et al. 2016; Jiménez-Marín et al. 2022; Nagy et al. 2014

R2-P1 · New genes at simple-to-complex steps?



Predicts Four unseen simple → complex events carry more sequence-novel families than matched controls.

Round-1 lead at nested events (1.9–3.3×); land-plant bursts; convergent expansions in complex fungi.

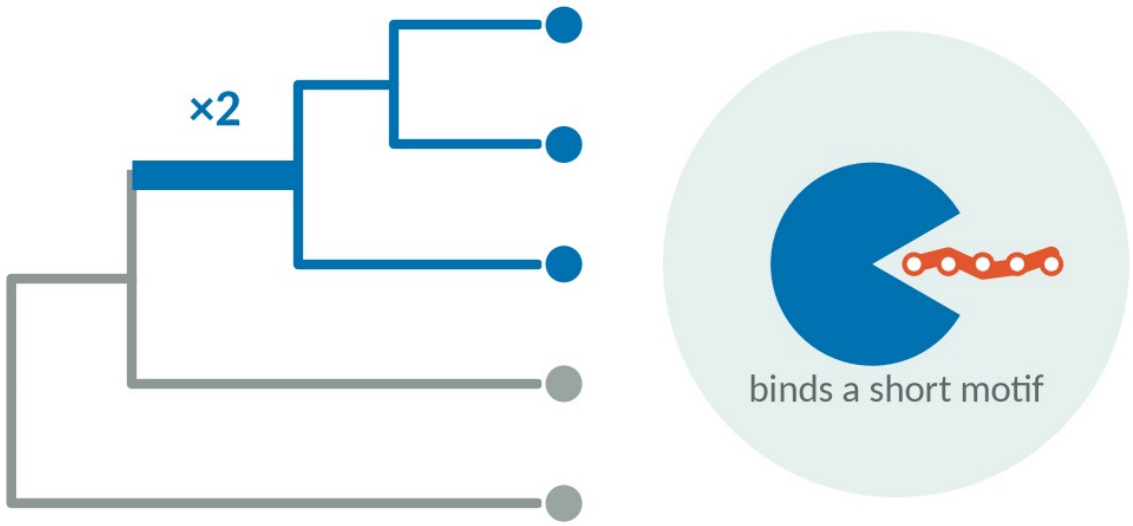
Power 18–41%; two events are single-species tips; three share one panel's controls.

NOT SIGNIFICANT

combined p = 0.050 (Holm 0.10); internal-only 2.8× [1.3–5.9], exploratory

Literature Knoll 2011; Bowles et al. 2020; Merényi et al. 2020; Nagy et al. 2018; Herron & Michod 2008; Lindsey et al. 2021

R2-P2 · More motif-binding domains?



Predicts Domains binding short linear motifs (SH2, SH3, PDZ, WW, 14-3-3…) are more abundant in multicellular genomes.

Proposed to expand with regulatory complexity; 115 families fixed from ELM before testing.

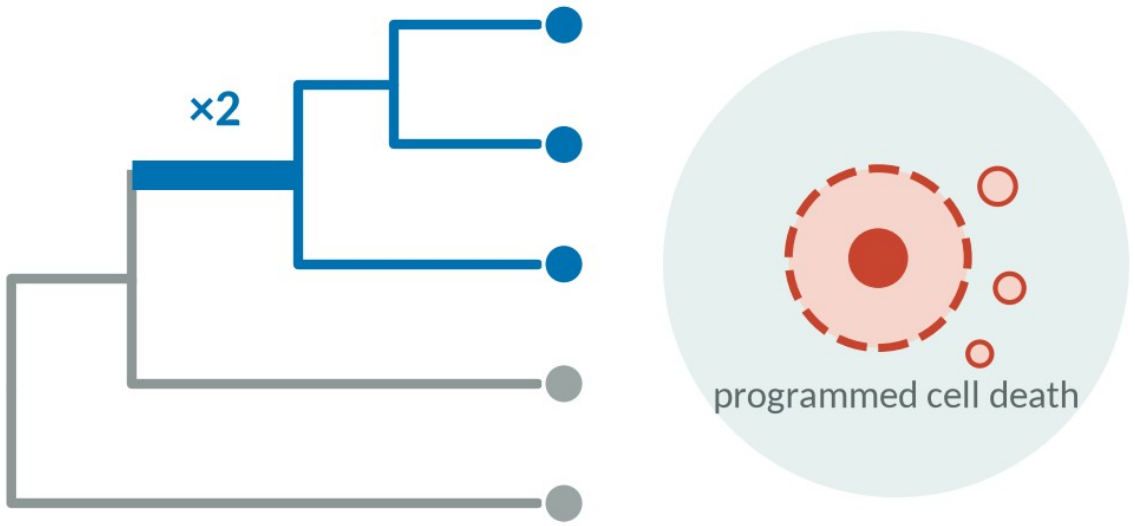
Expansion only within animals; contraction in hyphal fungi.

Literature Kumar et al. 2024 (ELM resource)

NULL

0.98-fold, p = 0.72; 6 of 7 leave-one-out estimates negative

R2-P3 · More cell-death machinery?



Predicts Multicellular genomes carry more programmed-cell-death proteins than expected for proteome size.

Cell death can benefit relatives; a joint origin with virus arms races has been proposed.

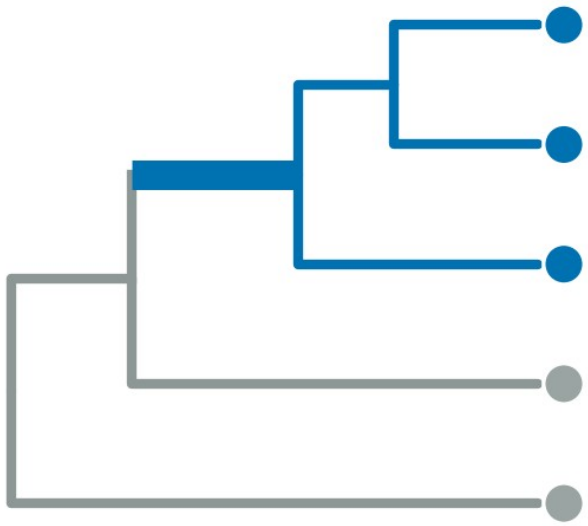
Size-dependent; mostly immune receptors; core apoptosis machinery already in the common ancestor of eukaryotes.

SUPPORTED

1.21× [1.04–1.42], Holm p = 0.014; mainly complex lineages

Literature Durand et al. 2016; Iranzo et al. 2014; Klim et al. 2018; Jones & Dangl 2006; Dyrka et al. 2014

R2 · Structural novelty and constraint





Predicts Families new at events lack structural relatives; structure changes faster per sequence change in multicellular clades.

Folds outlast sequences; ProstT5 + Foldseek make proteome-scale structural search feasible.

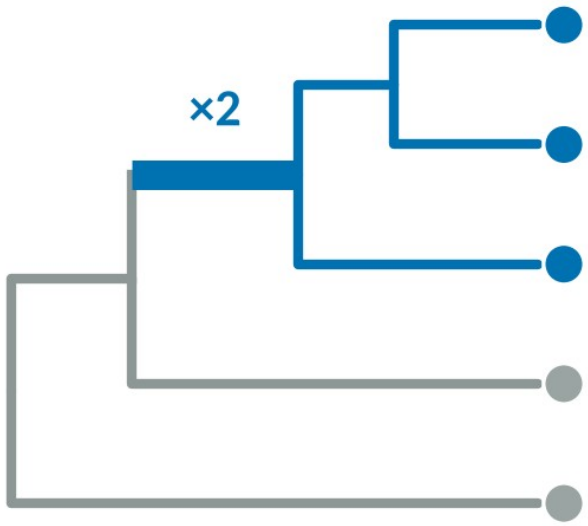
'No match' tracks short, disordered proteins; a shuffled-letter control gave the same 'effect'.

Literature Weisman et al. 2020; van Kempen et al. 2024; Heinzinger et al. 2024; Lynch & Conery 2003

OVERTURNED / INCONCLUSIVE

OR 1.44 → 1.18 [0.93–1.49] after adjustment; divergence test inconclusive

R2 · Protein complexes kept or duplicated?





Predicts Duplications on multicellular stems hit subunits of ancient complexes; complexes stay intact more often.

LECA interactome from co-fractionation mass spectrometry across 31 species.

Map holds only ancient complexes; 13 mappable species; duplication counts depend on the method.

Literature Cox et al. 2026 (Marcotte lab; preprint Cox et al. 2024)

NULL

66.6% (animals), 72.1% (fungi) vs 63.6% controls; none enriched after correction

H

Decisions that transfer

What this project did, what went wrong, and what future projects should copy



From this project to future projects

This project	Future projects
Hashed pre-registration, 3 tags	**Label tests with < 40% power as estimation**
Calibrated on simulations (2.1–2.6×)	**Pilot every statistic on control-only data**
Adversarial verification: 2 claims overturned	**Never let authors verify their own work**
Department-head review steered round 2	**External review at every round boundary**
Dense panels, 15–45 min each	**Many small runs beat one giant run**
EukaryOMA cross-check (29%)	**Run a second method on key counts**
GPU pilot caught a silent CPU fallback	**Check the device before long jobs**

Condensed from LESSONS.md (14 rules) and the fresh-agent retrospective.

What went wrong

Errata
3 round-1 overstatements; 8 mismatches in the final numbers audit

Overturned claims
structural novelty and stem agreement, after verification

Tool incidents
self-killing pattern kill (×2), orphaned GPU job, premature wait-loop run, undocumented floor

Low-power test
simple → complex test: 18–41% power, could never pass Holm

All were caught and documented by the process itself; none is hidden in the report.

Conclusion

AI-led research: outcome and future possibilities



AI-led research: honest, fast, self-correcting

≈31 h

two complete rounds, from literature to reviewed report

Not yet verified by a human

Did well

- Honest nulls, with power
- Self-correction: 2 own claims overturned
- Full audit trail: pre-registration, deviations
- Parallel team, cheap dense runs

Did badly

- Low-power confirmatory tests
- Design flaws found after seeing data
- Undocumented choices, stale numbers
- Tool handling, lagging records

Future possibilities

Human verification
methods and benchmarks, then update the disclaimer

More genomes
EukaryOMA's 1,328 genomes; genome sisters for transcriptome-only lineages

Deep-node orthology
domain-level dating; gene-birth simulator for the bias

Structure at scale
finish 3Di for all proteomes; real 3D models

Experimental links
cell death vs immunity; regulatory data

First step: a human check of methods and benchmarks; nothing here has been verified yet.

Read more

Final report: claude.ai/artifact/9rkyi2Xwe1VyhCgUcjZuoJ

Department-head review: claude.ai/artifact/NfrexbqRBFmS3A97VvTqcR

Added after publication: report/REPORT.md §12 · results/r2/recurrence_ladder/

Repository: report/REPORT.md · RESEARCH_TREE.md · LESSONS.md · reviews/

Hackathon: github.com/evocellnet/ai-hackathon · [repository URL: ____]

